

A Comprehensive Review: Analysis of Machine Learning, Deep Learning, and Large Language Model Techniques for Revolutionizing Driver Identification

Anwar Mehmood Sohail*, Ying Wah Teh*, Nauman Khan*, Zahid khan, and Anis Koubaa

Abstract: Driver identification is crucial for automotive security, law enforcement, and ride-sharing industries, as well as for advanced driver assistance systems, fleet management, and usage-based insurance. Machine learning and deep learning techniques show promise for accurate identification, but a comprehensive analysis of existing methods is lacking. This paper addresses the gap by reviewing and analyzing existing techniques, including preprocessing, feature extraction, classification algorithms, and deep learning architectures. Performance, advantages, and limitations are critically evaluated. A future framework for driver identification with Large Language Model (LLM) is proposed, exploring its potential in this domain. Traditional methods like Support Vector Machines and Random Forests offer reliable performance, while deep learning requires larger datasets and computational resources but achieves higher accuracy. Synergies between established techniques and emerging technologies like LLM are identified for future research. Key directions include hybrid approaches and transfer learning for efficient adaptation to new datasets. This review serves as a valuable resource for researchers and practitioners, highlighting strengths, weaknesses, and promising research directions for highly accurate driver identification systems.

Key words: Driver identification; driver recognition; machine learning; deep learning; large language models (LLM); review; ChatGPT; GPT

1 Introduction

Driver identification, also called driver fingerprinting, is the process of discerning the identity of the individual operating a vehicle^[1]. This process aims to recognize or verify the identity of a person who is

driving a vehicle, which involves the classification or identification of a driver. Human drivers exhibit a diverse range of behaviors while operating a vehicle^[2]. Consequently, this distinct driving behavior can serve as a digital fingerprint characterized by a unique style specific to each driver. Drivers demonstrate distinct

- Anwar Mehmood Sohail and Ying Wah Teh are with Department of Information Systems, Faculty of Computer Science and Information Technology, Universiti Malaya, Kuala Lumpur 50603, Malaysia. E-mail: s2042114@siswa.um.edu.my; tehyw@um.edu.my.
- Nauman Khan is with Robotics and Internet-of-Things (RIOTU) Laboratory, Prince Sultan University, Riyadh 11586, Saudi Arabia, and also with Department of CS & IT and AI, University of Malakand, Lower Dir 18800, Pakistan. E-mail: noumankhan@uom.edu.pk.
- Zahid khan is with Robotics and Internet-of-Things (RIOTU) Laboratory, Prince Sultan University, Riyadh 11586, Saudi Arabia. E-mail: zskhan@psu.edu.sa.
- Anis Koubaa is with Robotics and Internet-of-Things (RIOTU) Laboratory, Prince Sultan University, Riyadh 11586, Saudi Arabia, and also with College of Engineering and Advanced Computing, Alfaisal University, Riyadh 11533, Saudi Arabia. E-mail: akoubaa@alfaisal.edu.

* To whom correspondence should be addressed.

Manuscript received: 2024-03-11; revised: 2024-07-09; accepted: 2025-05-08

driving patterns influenced by their characteristics and skills^[3]. For instance, drivers may exhibit variations in their maximum and minimum speeds when traveling on the same road. Certain drivers may engage in frequent gas or brake pedal operations, particularly in heavy traffic or on uneven roads, while others may not exhibit the same behavior. Hence, these distinctive driving patterns can be effectively utilized to identify drivers^[4, 5, 6, 7]. Exploring this unique aspect of individual driving behavior style has been an area of extensive research in recent years and continues to be actively investigated. This research area has gained significant momentum, particularly with the increasing number of sensors and the abundance of data available in modern vehicles.

Fig. 1 illustrates the various domains in which this technology holds promise for numerous advantages, such as in the use of ride-sharing industry, automotive security, Advance driving assistance system (ADAS)^[8], usage-based insurance (UBI) systems by penalizing insurers who lend cars to other drivers, law enforcement agencies by confirming identity of

persons who break traffic laws, etc.^[9, 10, 11, 12, 13].

1.1 Automotive security

Vehicle transportation plays a crucial role in the lives of contemporary individuals. A significant portion of the population owns vehicles and utilizes them regularly. Nevertheless, there has been a notable rise in the occurrence of vehicle theft over the past decade. According to the FBI Uniform Crime Reports, the number of stolen vehicles has increased by 5% over the past five years. Additionally, statistics from the Home Office indicate that a staggering 111,999 vehicles were reported stolen between 2017 and 2018^[4].

Driver identification can be used to prevent unauthorized access to vehicles, protect against theft, and improve safety by ensuring that only authorized drivers operate the vehicle^[14, 15, 16]. In their study^[17], they introduce a driver authentication approach that utilizes the vehicle's sensors to analyze patterns of driver behavior. By employing Machine Learning (ML) algorithms, they generate individual driver profiles for the verification process, which are subsequently compared to existing profiles.

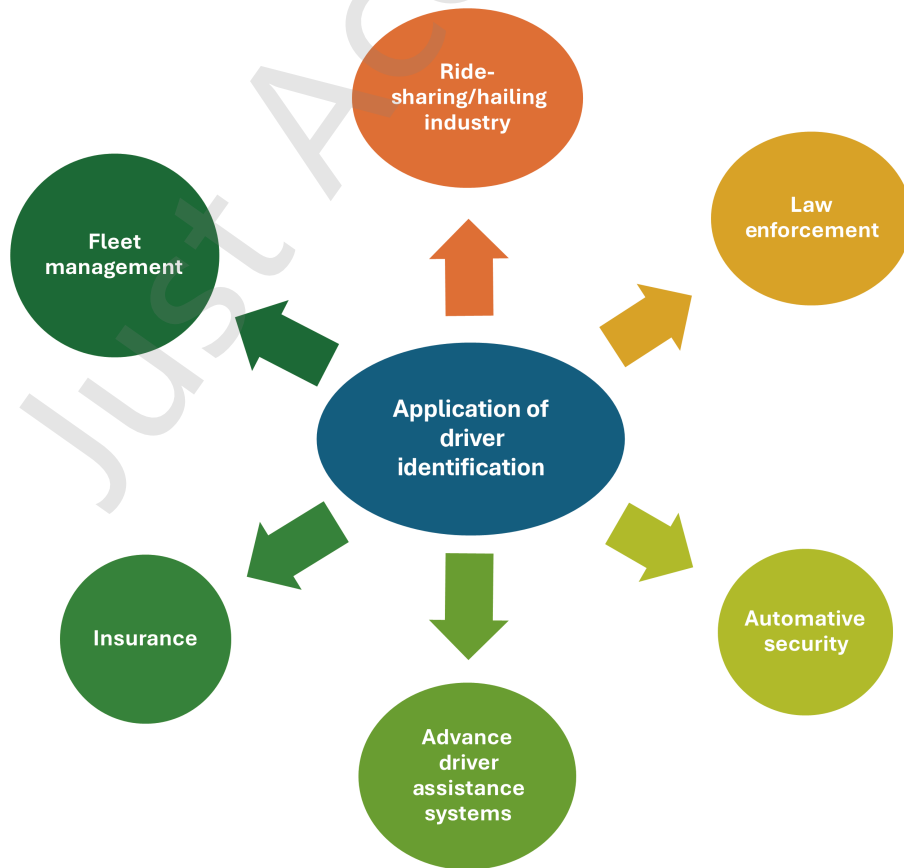


Fig. 1 Applications of Driver Identification for multiple stakeholders/sectors.

1.2 Law enforcement

Driver identification is crucial for law enforcement agencies to ensure road safety and prevent criminal activities involving vehicles. With driver identification technology, law enforcement agencies can accurately identify drivers involved in road accidents, traffic violations, and criminal activities and take appropriate legal action. The technology can also help agencies track stolen vehicles and identify suspects involved in hit-and-run cases^[18, 19, 20, 21, 22]. According to the research conducted in [19, 20], the utilization of in-vehicle digital data for driver identification within the realm of digital forensics holds the possibility for resolving hit-and-run accidents and other incidents involving vehicles.

1.3 Ride sharing/e-hailing industry

The transportation industry has undergone a significant transformation with the emergence of ride-sharing services, which cater to both professional drivers and the sharing of vehicles among multiple passengers. Driver identification is an important aspect of the ride-sharing industry, such as Uber or Cabify^[23, 24]. It helps to ensure the safety and security of passengers by verifying the driver's identity and ensuring that they are licensed and authorized to operate the vehicle. It also helps to prevent fraudulent activity by ensuring that only authorized drivers can provide rides through the platform^[25].

1.4 Advanced Driver Assistance Systems (ADAS)

Driver identification can also play a vital role in ADAS by personalizing the system settings based on the driver's preferences and driving behavior^[26]. For example, once the driver is recognized, the ADAS can automatically adjust the seat position, mirror angles, and steering sensitivity^[27, 28]. This can improve the overall driving experience and provide greater comfort for drivers. In their study^[29], tackle driver identification within infotainment. They propose that, depending on who is driving, the ADAS in the car may be customized to accommodate particular family members.

1.5 Insurance

Driver identification is crucial for insurance companies to accurately assess the risk posed by each driver and determine the appropriate insurance premiums. By analyzing data about the driving behavior of each

driver, insurance companies can provide more personalized insurance plans and reward safe driving practices, leading to fairer and accurate pricing for customers. Additionally, driver identification can help prevent fraudulent claims and reduce insurance fraud, ultimately benefiting both insurance companies and their customers^[18].

1.6 Fleet management

Driver identification plays a crucial role in fleet management by allowing companies to monitor and track the driving behavior of individual drivers. This information can be used to optimize route planning, monitor fuel consumption, and reduce vehicle wear and tear, ultimately enhancing efficiency and resulting in cost savings^[30, 31].

Previous studies in driver identification have established a standardized framework called the driver identification pipeline, which outlines the procedures involved in designing and implementing driver identification systems. This pipeline encompasses various stages, including driving data collection, data preprocessing, feature extraction, dimensionality reduction, building classification models, as well as testing and validating the learning models. Figure 3. illustrates the process.

The driver identification process initiates with the collection of data, as illustrated in Figure 3. This involves gathering various types of data sources, such as biometric, behavioral, and contextual data about the driver using sensors, cameras, and other devices. These data sources are discussed in detail in Section 2. Data pre-processing is another crucial step, which entails transforming the raw driving data into a suitable representation. The pre-processing begins with filtering to remove noise^[32, 33, 34, 35, 4], data cleaning^[36, 37, 38], and feature scaling^[39, 38, 4].

Time windowing divides the data into smaller segments using either overlapping or fixed window sizes, enabling the extraction of distinctive features from each segment^[40, 41]. Class balancing techniques address class imbalance^[21], and encoding categorical variables^[42, 43]. This comprehensive pre-processing pipeline enhances data quality, ensures consistent scaling, captures temporal patterns^[44, 45], increases diversity, addresses class imbalance, and enables effective numerical representation, leading to accurate and reliable driver identification models^[46, 47]. A comprehensive discussion of these pre-processing

methods can be found in Section 4, where they are examined in detail.

To create a useful collection of features, feature extraction and feature selection approaches are used, with the goals of improving accuracy, speeding up computation, and reducing classification mistakes. There are shallow features and deep features in the process of obtaining driving features. Shallow features include the extraction of traditional handmade characteristics, such as statistical time-domain and frequency-domain features. However, shallow features heavily rely on domain expertise, necessitate a substantial amount of labeled driving data, and employment of dimensionality reduction techniques that lack generalizability. Deep learning (DL) is a successful solution to address the aforementioned challenges due to its capability to construct features effectively.

DL algorithms are end-to-end ML algorithms that can be trained on raw data to automatically learn the representations needed for tasks like detection or classification. In various fields, DL methods outperform traditional approaches, particularly in constructing high-dimensional representations for complex data. The most significant advantage of DL is its ability to automatically map low-dimensional, shallow features to high-dimensional, deep features based on the objective function, eliminating the need for manual feature engineering or domain expertise.

Since 2016, there has been a shift towards DL algorithms replacing conventional ML techniques for feature extraction and their use in driver identification tasks. The study^[48] introduced various Deep Neural Network (DNN) architectures, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), to learn representations of drivers' driving styles. After that, several other studies utilized DL methods and indicated that this approach outperformed traditional ML algorithms. DL techniques are particularly popular in pattern recognition, image processing, speaker recognition, natural language processing, and now driver identification. Examples include deep autoencoder (DAE), CNN, and RNN. These techniques and their respective studies are discussed in detail in Section 7.

Driving features combined with ML algorithms to build a driver identification model. Some of the ML algorithms employed in this context include Support Vector Machine (SVM)^[49, 50, 51], Decision Tree

(DT)^[52, 31], and Random Forest (RF)^[19, 20]. The specifics of these classification algorithms are discussed in Section 6. In the case of DL, both feature extraction and classification are trained as part of the model construction process. In the end, the driver identification system undergoes evaluation using diverse performance metrics, including accuracy, precision, and recall. These metrics help assess the effectiveness and reliability of the system in accurately identifying drivers.

The goal of this work is to undertake a thorough analysis of driver identification systems that make use of various DL and ML classifiers. In this paper, we seek to provide a comprehensive summary of current developments in driver identification. We go through the datasets and data sources with their characteristics, several driving pre-processing strategies, and feature selection techniques in particular. We also look at several ML classifiers and DL techniques used in building driver identification models, as well as assessment metrics to rate the effectiveness of classification methods. Furthermore, this study discusses the potential use of Large Language Model (LLM), which has already covered many fields such as medical^[53], engineering^[54, 55], and so many. and future research directions for further enhancements and areas of focus based on the findings of the reviewed studies. Figure 2. illustrates the taxonomy used in this review.

2 Related work

Several surveys and review studies on driver identification have been published recently; some of them that are closely related to our survey are shown in Table 1. However, these articles primarily focus on different ML and DL methods and techniques used for driver identification. In contrast, this study not only emphasizes ML and DL methods but also delves into datasets, data sources, pre-processing methods, feature selection techniques, and the potential use of LLMs for driver identification.

A closely related review was conducted by^[56], which examined various data sources, ML, and DL algorithms for driver identification. Their work provided a broad overview of the field but did not thoroughly explore the integration of LLMs or the latest advancements in DL algorithms. Our work addresses these aspects by providing an in-depth examination of significant new investigations into cutting-edge DL algorithms for driver detection and exploring the emerging role of

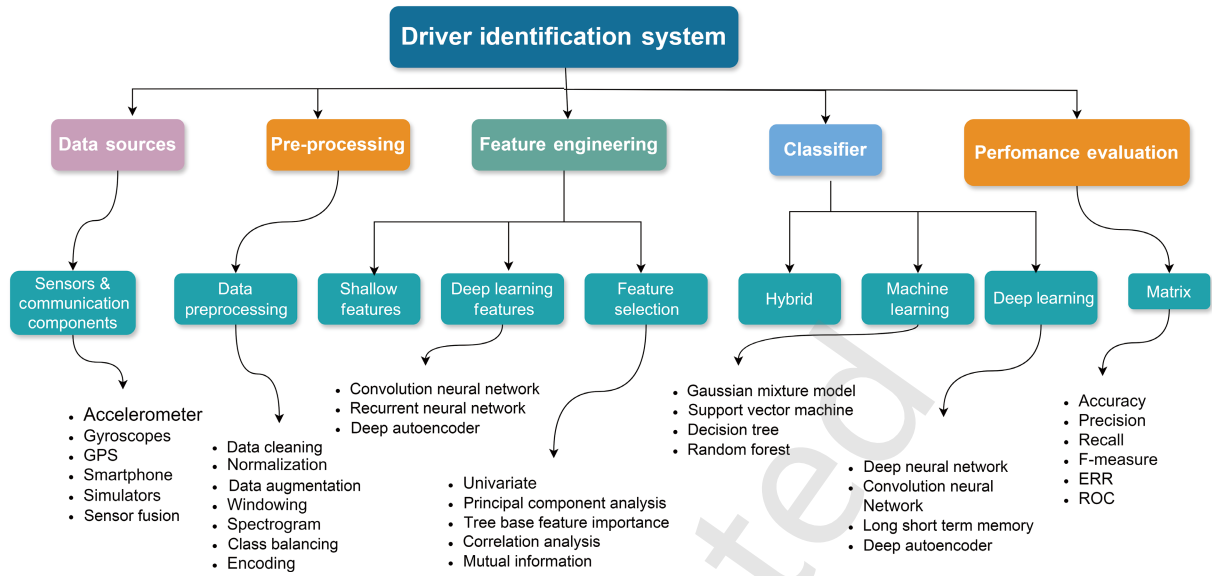


Fig. 2 Taxonomy of driver identification system.

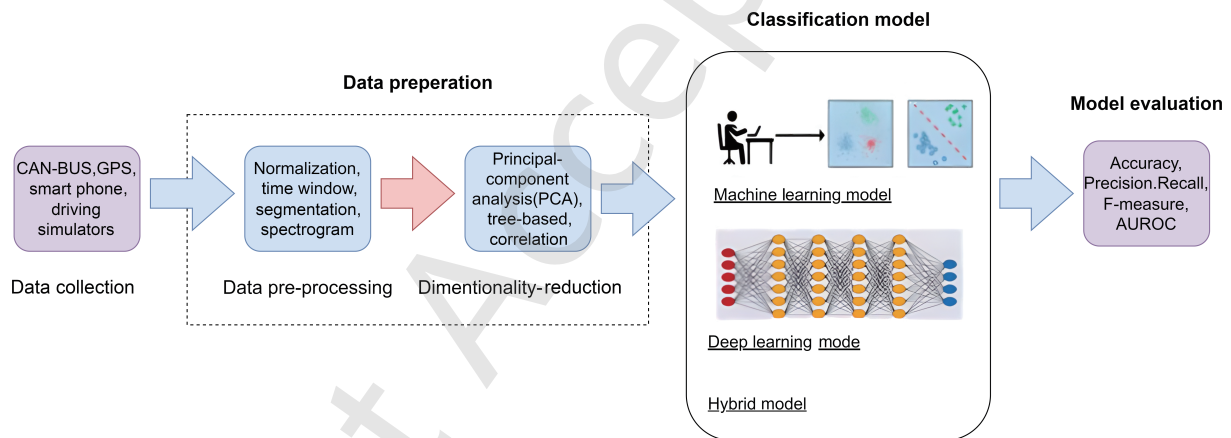


Fig. 3 Classical driver identification process.

Table 1 Comparison of recent studies on driver identification. The comparison is based on inclusion (✓) and exclusion (×) across preprocessing methods (Prep), data sources (DS), datasets (DST), feature selection (FSel), ML, DL, LLM, and evaluation metrics (EMetr).

Publication	Year	Prep	DS	DST	FSel	ML	DL	LLM	EMetr
"Driver Identification Methods in Electric Vehicles, a Review" ^[56]	2022	×	✓	×	×	✓	✓	×	×
"Driving Behavior Analysis Guidelines for Intelligent Transportation Systems" ^[57]	2022	×	✓	✓	×	✓	✓	×	×
"Machine Learning for Security in Vehicular Networks: A Comprehensive Survey" ^[58]	2022	×	×	×	×	✓	×	×	×
<i>This Study</i>	2024	✓	✓	✓	✓	✓	✓	✓	✓

LLMs in this domain.

For instance, the study^[57] compared and analyzed different approaches for driver identification based solely on driving behavior. They focused on behavioral features such as acceleration patterns and braking behaviors to identify drivers. However, their study did

not consider physical or physiological features, which can be crucial for enhancing identification accuracy. Our review addresses this gap by incorporating a broader range of data sources, including physiological signals and physical attributes, thereby providing a comprehensive analysis of multimodal data for driver

identification.

Similarly, the study^[58] presented a review that specifically discussed traditional ML algorithms used for driver identification or fingerprinting. Their study concentrated solely on traditional ML algorithms, providing an overview of methods such as SVMs and DTs. However, they lacked detailed explanations of recent studies that propose various DL algorithms. This limitation underscores the necessity of our review, which encompasses both traditional and cutting-edge DL techniques, offering a more holistic view of the advancements in this field.

In summary, while previous studies have laid the groundwork in driver identification through traditional ML and DL methods, our review expands on these foundations by incorporating a broader array of techniques. This includes the latest advancements in DL and the innovative application of LLMs, aiming to revolutionize the field of driver identification.

2.1 Motivation and Contribution

The present study differs from previous reviews in several aspects. Firstly, it encompasses the most up-to-date techniques and methods employed in driver identification, providing a comprehensive analysis of different ML and DL techniques utilized for this purpose. It also examines the various data sources employed in driver identification. Secondly, this review goes beyond the scope of previous studies by including datasets, pre-processing techniques, feature selection methods, and LLMs specific to driver identification. There is no existing comprehensive review available, that addresses all the components necessary for implementing a driver identification system to the best of our knowledge. A comparison of recently released survey papers is shown in Table. 1, highlighting the areas they cover for reference. The paper makes the following contributions:

- Summarizing recent advancements in the construction of datasets, pre-processing, feature engineering, and the utilization of a variety of ML classifiers and DL methods for driver identification.
- Providing an analysis of driver identification techniques, outlining their strengths and weaknesses.
- Proposed a future framework for driver identification with LLM and highlighting applications of the advanced LLM framework, and its potential to revolutionize this field.
- Emphasizing future challenges in the creation of

datasets, feature engineering, classification models for driver identification, and identifying research gaps in these areas.

The remainder of this research is structured as follows: A review of data sources and datasets commonly used for driver identification is presented in Section 3. Data pre-processing techniques are discussed in Section 4. Various feature selection methods are explored in Section 5. ML and DL methods for driver identification are covered in Sections 6 and 7, respectively, highlighting their respective strengths and weaknesses. Section 8 and 9 describes the framework and utilization of LLM in the context of driver identification, respectively. Section 10 details the performance assessment of ML and DL models for driver identification. Section 11 addresses open issues, observations, and future research directions for further improvements and focus. Finally, this comprehensive revised article is concluded in Section 13.

3 Data sources for Driver Identification

In the domain of driver identification, researchers and practitioners have extensively explored a wide array of data sources to develop robust and accurate identification techniques^[59]. This section delves into the exploration of numerous data sources used for driver identification, aiming to provide insights into the diverse range of information that can be harnessed for this purpose. Recognizing the significance of driver identification in various applications, including vehicle security, personalized driver assistance, and insurance risk assessment, the selection and analysis of appropriate data sources are crucial steps in the development of effective identification methods. Additionally, the strengths and weaknesses of each data source are examined, shedding light on their respective advantages and limitations in the context of driver identification. By comprehensively exploring and understanding these data sources, researchers and practitioners can leverage their unique characteristics to develop more accurate and reliable driver identification systems, contributing to the advancement of driver-centric technologies and systems. Fig. 4 and Fig. 5 show the frequency of datasets and data sources used in the literature, respectively.

Some researchers utilize driver biometric data for driver identification purposes^[60]. This includes data such as facial features^[61, 24], actions^[62], fingerprints^[63],

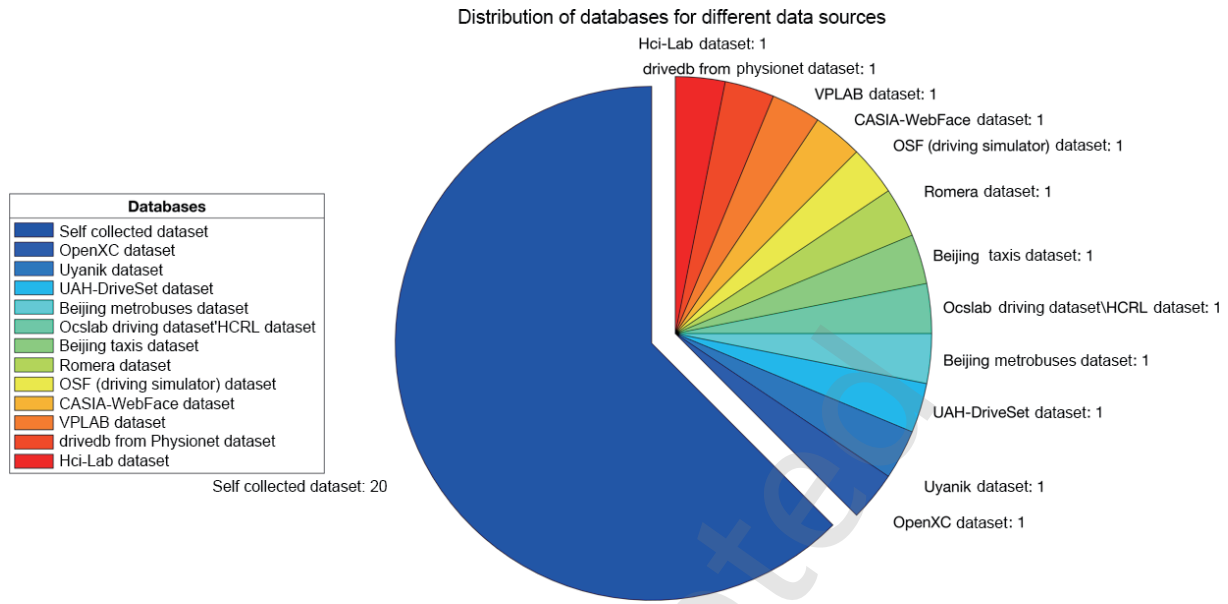


Fig. 4 Distribution of dataset for driver identification system from multiple sources.

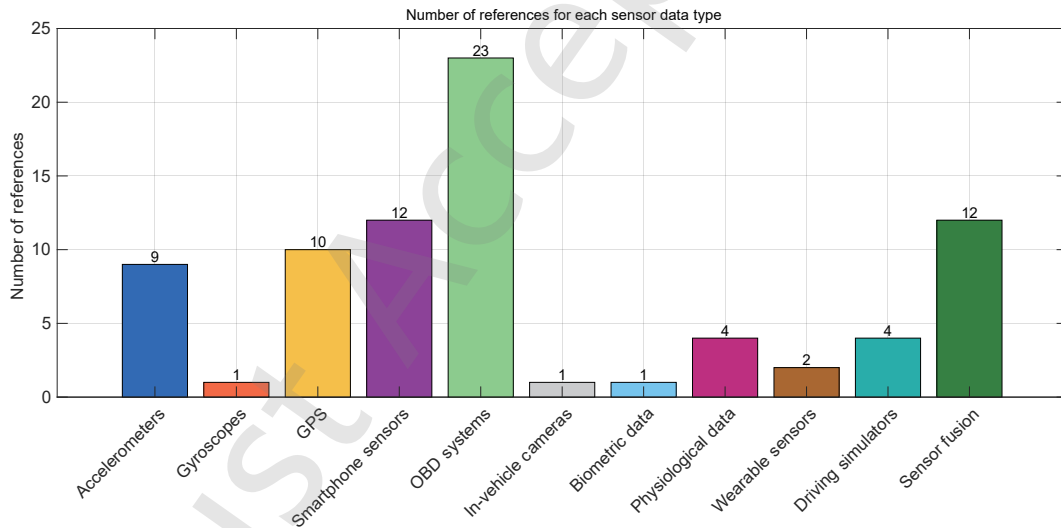


Fig. 5 Frequency of various data sources for driver identification used in the reviewed studies.

voice characteristics^[64], physiological states^[65], gestures^[66], and grip patterns^[67]. These kinds of data provide significant relevance and excellent identification accuracy since they directly represent the distinctive biological traits of the driver. However, challenges arise in acquiring such data, as it often requires new devices, leading to issues like space requirements and increased costs. Additionally, continuous real-time recognition poses difficulties, and there are concerns regarding the collection of information that may infringe on personal privacy. Furthermore, limitations in application arise, including the potential for data falsification. Therefore, we

reviewed all the data sources that are recently emerging for driver identification, despite one study which is based on facial features^[24].

3.1 Smartphone sensors

Smartphone sensors have emerged as a valuable and accessible data source for driver identification^[10, 16, 24]. With their increasing ubiquity, smartphones offer a range of built-in sensors, including accelerometers, gyroscopes, Global Positioning System (GPS), and magnetometers, which can capture various aspects of driver behavior and vehicle dynamics. These sensors enable researchers to collect large-scale and real-world

driving data for identification purposes^[29, 30, 41, 43]. The strengths of smartphone sensors lie in their convenience, as most individuals carry smartphones with them while driving, allowing for continuous data collection without the need for additional devices^[49, 68, 69]. Additionally, they make it possible to acquire data from vehicles that do not support Controller Area Network (CAN) bus data. Moreover, smartphones offer a diverse set of sensors that can capture different dimensions of driving behavior^[70, 71, 72].

However, smartphone sensors also have certain limitations. One key weakness is the potential variability in sensor quality and accuracy across different smartphone models, leading to potential inconsistencies in data collection and reliability. Additionally, issues related to sensor calibration, battery life, and data privacy need to be addressed when utilizing smartphone sensors for driver identification. Despite these challenges, the use of smartphone sensors holds great potential for driver identification, offering a scalable and cost-effective approach to gather valuable data on driver behavior and patterns.

3.1.1 Accelerometer and Gyroscope

Several studies have been conducted to explore accelerometers and gyroscope sensors for driver identification^[73, 68, 72, 41] measured driving behavior signals from a smartphone and accurately identified three drivers using only two acceleration signals. In study^[18], a driver identification system relying on histograms of acceleration data acquired from the accelerometer sensor achieved an accuracy of up to 99% and demonstrated effective imposter detection. However, it is important to note that the dataset utilized in their research was confined to campus shuttle buses, limiting the generalizability of their findings. The study^[69] also investigated driver identification using a 3-axis accelerometer and found it performed best. However, when there were more drivers, the algorithm's performance suffered.

A driver recognition system that made use of drivers' maneuvers^[74, 75, 76], such as right and left turns, was reported in [51], which utilized acceleration and orientation data. More recently, the study^[41] proposed a driver identification system based on smartphone accelerometer and gyroscope signals, while another^[16] demonstrated competitive results using features extracted from smartphone accelerometers. Accelerometers and gyroscopes serve as valuable

alternatives in environments with tall buildings and tunnels, where GPS receivers frequently encounter difficulties. These sensors prove especially useful in such scenarios, providing reliable data even when GPS signals are unavailable. Additionally, using accelerometers and gyroscopes mitigates concerns regarding privacy invasions that are frequently connected to GPS data.

3.1.2 Global Positioning System (GPS)

GPS is widely used for driver identification, providing valuable insights into driver behavior and patterns^[77, 78, 29, 72]. It accurately tracks vehicle location, movement, and spatial behaviors. The strengths of GPS include precise trajectory information, real-time data availability, and analysis of driving patterns^[10, 31]. However, GPS has limitations such as signal interference, satellite availability, and limited granularity for fine-grained identification^[42, 71]. Nonetheless, GPS remains a valuable tool for studying driver behavior on a larger scale and can be combined with other data sources to enhance identification accuracy and reliability.

Several researchers have used GPS for driver identification, showcasing its feasibility and effectiveness. Study^[70] demonstrated the potential of smartphone GPS data for distinguishing drivers and achieved an accuracy of 82%. Another study^[49, 79] created a method for identifying drivers that was based on GPS-derived driving patterns. This system achieved a remarkable accuracy of 96%.

3.1.3 CAN-BUS/OBD-II

CAN-BUS/OBD-II data has been extensively utilized for driver identification purposes in various research studies^[80, 71, 81, 82, 83]. The utilization of CAN-BUS data enables the extraction of valuable information related to vehicle parameters and behavior, offering insights into driver identification^[84, 4, 13, 14, 23, 15]. One of the notable strengths of CAN-BUS data is its availability in most modern vehicles, making it a convenient and accessible source for driver identification^[28, 37, 38, 85]. Additionally, the use of CAN-BUS data allows for cost-effective solutions as it eliminates the need for additional sensors or devices^[42, 43, 52]. However, it is essential to acknowledge some limitations of CAN-BUS data for driver identification^[86, 87, 6]. One key weakness is the potential lack of standardization across different vehicle models, which may affect the consistency and compatibility of the data^[88]. The interpretation and

analysis of the significant amount of complicated data produced by the CAN-BUS presents another difficulty, requiring sophisticated algorithms and techniques for accurate driver identification.

Several researchers have explored the use of CAN bus data for driver identification across a variety of applications. In the context of infotainment systems, CAN-bus data was leveraged to address driver identification challenges^[29, 89]. The potential of CAN-BUS data for digital forensics, particularly in hit-and-run accident scenarios, has also been investigated^[19, 20]. A ML-based approach for driver profiling using in-vehicle CAN-BUS sensor data was proposed^[52]. Another study introduced a DL system that relies solely on steering behavior extracted from CAN-BUS data^[32]. Additionally, CAN bus signals have been used to accurately predict driver identity^[21].

3.1.4 Biometric data

Biometric data has emerged as a promising approach for driver identification, leveraging individuals' unique physiological or behavioral characteristics. Biometric data sources such as fingerprint, voice, iris, and facial recognition^[24] have been employed in driver identification systems. Biometric data's strengths lie in their uniqueness, making them highly reliable for individual identification. They provide a non-intrusive and convenient means of identification, requiring minimal effort from the driver. Additionally, biometric data can be difficult to forge or manipulate, enhancing security. However, biometric data also has its limitations. Environmental influences, sensor limitations, and human variability can all impact the reliability and accuracy of biometric data. Biometric systems may also face privacy concerns and potential biases. Using facial recognition, the study^[24] provides a real-time and reliable driver identification approach. Introducing a lightweight CNN model, they achieved an average verification accuracy of 87.44%.

3.1.5 Physiological sensors

Physiological sensors, such as Electrocardiogram (ECG) and Electromyogram (EMG), have been explored for driver identification due to their ability to capture unique physiological signals^[27, 36, 71, 90]. The strengths of physiological sensors lie in their high specificity and reliability, as they directly reflect the individual's physiological responses. They can offer real-time monitoring of the driver's physiological state, allowing for immediate identification of anomalies or

changes in driver behavior. However, physiological sensors also have some limitations. They require direct contact with the driver's body, which can be uncomfortable or inconvenient in certain situations.

The accuracy and effectiveness of physiological sensors can be influenced by sensor placement, signal interference, and individual variations in physiological responses. Additionally, using physiological sensors for driver identification may raise privacy concerns related to collecting and storing sensitive health data. Therefore, while physiological sensors offer the potential for driver identification, carefully considering user comfort, signal quality, and privacy safeguards is necessary for their successful implementation. A driver identification system based on ECG morphological features was proposed^[27, 36]. It has also been demonstrated that incorporating driver-related information such as heart rate and driving behavior data can significantly enhance the effectiveness of driver identification^[71].

3.1.6 Smartwatch

Smartwatches have been utilized for driver identification, leveraging their built-in sensors such as accelerometers and heart rate monitors. The strengths of smartwatches include portability, convenience, and continuous monitoring. They provide valuable data for driver behavior analysis. However, limitations include limited sensor accuracy, dependency on user compliance, and potential privacy concerns due to continuous data collection. In their study^[51], introduced the concept of driver identification using smartwatches. Their system concentrated on analyzing drivers' maneuvers, specifically right and left turns, utilizing data gathered from smartwatches. The system underwent evaluation in both real-world and simulated environments, yielding an error rate of around 18%. This research highlights the potential of smartwatches as a means of driver identification.

3.1.7 Driving simulators

Driving simulators offer researchers a controlled setting to gather data on driver behavior and performance^[91]. These simulators can capture various types of data, including vehicle dynamics, driver inputs (such as steering, acceleration, and braking), eye movement data (eye tracking), and physiological responses (heart rate, skin conductance, etc.). While driving simulators provide flexibility in simulating diverse driving conditions, they have limitations in

replicating the exact real-world driving experience.

Studies have shown differences between simulator-derived data and data collected in real-world scenarios^[40]. Nevertheless, researchers have successfully applied ML techniques, such as Gaussian Mixture Model (GMM), Hidden Markov Model (HMM)^[92], and SVM, on simulated driving data to achieve recognition accuracies ranging from 81% to 95%. For instance, one study reported an 81% recognition accuracy using GMM^[93], while another improved it to 89.6%^[94], and a further study achieved 95% accuracy using SVM^[95]. However, it is essential to acknowledge the limitations, such as differences between simulator data and real-world driving data, which may impact the generalizability of the findings.

3.1.8 Sensor fusion

Sensor fusion techniques have been employed for driver identification, combining data from multiple sensors to enhance accuracy and reliability^[10, 29, 36]. Sensor fusion can provide a comprehensive view of driver behavior by integrating information from various sources such as accelerometers, gyroscopes, GPS, and cameras^[41, 42]. This approach allows for a more robust identification system that considers multiple modalities simultaneously^[43, 71, 72]. The strengths of sensor fusion include improved accuracy through data integration, better contextual understanding of driver actions, and the ability to capture a wide range of driving features. However, challenges in sensor calibration, synchronization, and data fusion algorithms can pose weaknesses, requiring careful design and implementation to ensure optimal performance.

Many studies combine multiple data sources to enhance the accuracy of driver identification. One approach utilized signals from both GPS devices and the CAN bus to identify drivers^[29]. Similarly, accelerometer and GPS signals were used to develop a driver identification system that achieved 81% accuracy with 13-second-long test data, regardless of route or traffic conditions^[72]. A method to optimize driver detection using data from GPS and in-car sensors has also been proposed^[42]. To further improve identification accuracy, a system that simultaneously uses ECG and EMG signals in various driving states—reflecting real driving environments—has been suggested^[36]. The results indicated that combining EMG and ECG data yields better accuracy than using either signal alone. The performance of an optimized

DL model integrating smartphone and CAN-BUS data was evaluated^[43]. Additionally, a driver identification system based on smartphone accelerometer and gyroscope signals has been proposed^[41].

Table 2 summarizes the wide array of data sources while Table 3 shows the datasets used by the researchers and practitioners in the domain of driver identification.

4 Data pre-processing for driver Identification

In this section, we discussed the data pre-processing techniques (Table 4) utilized for driver identification. We highlighted the strengths and weaknesses of these techniques as outlined in the papers. These pre-processing methods have been carefully selected and evaluated to enhance the effectiveness and accuracy of driver identification models. By understanding their strengths and weaknesses, we can gain insights into their suitability for different datasets and contexts, ultimately informing the development of robust driver identification systems.

4.1 Data Cleaning

Data cleaning involves handling missing values, outliers, irrelevant features, noise, and periods of 0 km/h speed, such as when vehicles stop at traffic signals or in traffic congestion. This process is crucial for ensuring that subsequent analyses are based on reliable and accurate data. One study removed inactive periods (0 km/h) associated with bus stops from the dataset and demonstrated that applying an inactive-period filtering module significantly improved accuracy—by as much as 18%^[18].

4.2 Feature Scaling

Feature scaling is crucial to ensure that the input features are standardized to a similar scale. It prevents certain features from dominating others and helps in achieving better convergence during model training. Common scaling methods for feature scaling include normalization techniques such as min-max scaling and standardization techniques such as z-score scaling.

4.3 Data augmentation

Data augmentation is a method used to artificially increase the quantity and variety of the training sample. In the realm of driver identification, data augmentation

Table 2 Identified strengths and weaknesses of various types of data sources used for driver identification.

Data source	Strengths	Weaknesses	References
Accelerometers	provide information on vehicle dynamics and motion. Captures acceleration, deceleration, and lateral movement.	Limited to vehicle-centric information. Requires accurate calibration and placement for reliable measurements. May not capture nuanced driver behavior or contextual information.	[16], [18], [41], [51], [68], [69], [72]
Gyroscopes	Measures the rotational motion and orientation of the vehicle. Provides insights into steering control and maneuvering.	Limited to vehicle-centric information. Requires accurate calibration and placement for reliable measurements. May not capture driver-specific behavior or characteristics.	[41]
GPS	Provides accurate vehicle location and trajectory information. Captures driving patterns, routes, and speed.	May have limitations in accuracy and precision, especially in dense urban areas or areas with poor GPS reception. Requires a clear line of sight to satellites for reliable data collection.	[10], [29], [31], [42], [49], [70], [71], [72], [79]
Smartphone	Wide availability and high adoption rate of smartphones. Captures driving-related data (speed, location) and audio cues. Potential for integrating multiple sensors (accelerometers, GPS, microphones). Non-intrusive data collection method.	Limited to capturing specific aspects of driving behavior. Challenges in accuracy and consistency of sensor measurements. Privacy concerns related to collecting personal smartphone data. May raise concerns of driver distraction if not used appropriately.	[10], [16], [24], [29], [30], [41], [43], [49], [68], [69], [70], [71], [72]
CAN-BUS/OBD-II	Access to real-time vehicle performance data. Standardized data format across different vehicles. Wide availability in modern vehicles. Provides insights into driving behavior (speed, fuel consumption, braking).	Limited driver-specific information. Data may not capture nuanced driving behavior or contextual information. Limited to vehicles equipped with OBD systems (older vehicles may not have this capability).	[4], [10], [13], [14], [15], [19], [20], [21], [23], [28], [29], [31], [32], [37], [38], [42], [43], [52], [71], [81], [82], [83], [85], [88], [86], [87], [89]
Biometric Data (Camera)	Unique to individuals, enabling reliable identification. High visual data resolution. Comprehensive view of the driver. Ability to capture facial expressions. Can capture driver actions and gestures. Difficult to manipulate or impersonate. Non-intrusive data collection method.	Requires specialized sensors or systems for data collection. Privacy concerns and ethical considerations. Challenges in data processing and analysis due to variability in lighting conditions, occlusions, and facial recognition algorithms. Limited in capturing dynamic changes in biometric characteristics (e.g., appearance changes over time).	[24]
Physiological Data (ECG&EMG)	Provides physiological responses related to stress, arousal, and cognitive states. Provides insights into driver conditions.	Requires specialized sensors and equipment for data collection. Challenges in calibration and accuracy of physiological measurements. Privacy concerns regarding collection of personal physiological data. Intrusive.	[27], [36], [71], [90]
Smartwatch	Convenient and non-intrusive data source. Portability and ease of use. Ability to collect physiological data. Integration with existing smart devices.	Limited sensor capabilities. Limited battery life for continuous usage. Limited compatibility with other devices. Limited accuracy for certain measurements.	[51]
Driving Simulators	Controlled environment for data collection. Ability to recreate various driving scenarios. Simultaneous recording of multiple data types (vehicle dynamics, driver inputs, eye movements, physiological responses).	Lack of real-world driving context. Possible influence of simulator-induced effects on driver behavior. Limited external validity (findings may not generalize to real-world driving).	[10], [40], [51], [71], [91], [92], [93], [94], [95]
Sensor Fusion	Improved accuracy through data integration. Robustness against individual sensor failures. Enhanced discrimination of driver characteristics. Comprehensive understanding of driver behavior.	Increased complexity in system design. Reliance on multiple sensor calibration. Potential privacy concerns with multi-modal data. Cost and maintenance of multiple sensors.	[10], [29], [36], [41], [42], [43], [71], [72]

Table 3 Detail description of datasets for driver identification which is used in reviewed studies.

Dataset	Available features/sources	Detail of Dataset	References
Ocslab Driving dataset/HCRL Dataset (Kwak, Woo et al. 2016) ^[17]	CAN-BUS/OBD-II Data	No of Drivers: 10, Avg trip: 46km, 23hr, 2 trips/driver, Road type: City ways, motorways, and parking spaces, Time: 8 pm - 11 pm on weekdays, Record/size: 94,401, No of Features: 51, Sampling rate of 1 Hz	[4], [10], [13], [20], [28], [32], [38], [43], [52], [71], [82], [85], [87], [88], [89]
OpennXC (Ford Bug Labs 2020) ^[6]	CAN-BUS/OBD-II Data	No of Drivers: 3, No of Features: 7, Road type: Freeway and highway	[4]
(Romera, Bergasa et al. 2015) ^[5]	CAN-BUS/OBD-II Data	No of Drivers: 6, No of Features: 7, Road type: Motorway and secondary road	[4]
Uyanik dataset (Takeda, Hansen et al. 2011) ^[22]	CAN-BUS/OBD-II Data, GPS, gyroscope, video camera, infrared range sensors	No of Drivers: 105, Gender: 17 female and 88 male, Avg trip: 25-km route of Istanbul, Record/Size: 9TB	[21]
VPLAB Dataset (Abut, Erdoğan et al. 2007) ^[35]	CAN-bus, IMU, GPS, cameras, and laser scanners	No of Drivers: 95, Avg trip: 25km/47:68 m, Road Type: City ways, motorways, and highways, Record/Size: 9 TB	[16], [32]
UAH-DriveSet Dataset (Romera, Bergasa et al. 2016) ^[12]	Smartphone sensors (GPS, camera, internet, internal sensors)	No of Drivers: 6, Age: 20-50 years, Avg trip: 500 minutes, 25km, with each trip: 7, Road type: Motorway, secondary roads, Freq: 10hz, Speed: 90-120km/hr	[10], [71]
Hci-Lab Dataset (Schneegass, Pflöging et al. 2013) ^[34]	GPS + physiological + camera + smartphone	No of Drivers: 10, Gender: 7 male, 3 females, Ages 23 – 57, average of 35 years old, Road Type: Highways and tunnels, Record/Size: 2.5 million, 450 MB	[10], [32], [49], [71]
OSF (driving simulator) (Taamneh, Tsiamyrtzis et al. 2017) ^[11]	GPS + physiological + camera	No of Drivers: 68, Each trip: 8 Freq: 1hz	[10]
drivedb from Physionet's databank	ECG + EMG (Physiological)	Link: PhysioBank ATM (physionet.org)	[36], [90]
CASIA-WebFace Dataset (Yi, Lei et al. 2014) ^[33]	Images	No of Drivers: 10575, Record/Size: 494 414 images	[24]
Beijing Taxis Dataset (Yu, Zhao et al. 2010) ^[50]	GPS	No of Drivers: 8500+, Avg trip: 1 month, Road Type: 42% of all taxis in the urban area of Beijing	[49], [79]
Beijing Metrobuses Dataset (Rahim, Zhu et al. 2020) ^[49]	GPS	No of Drivers: 10 city bus drivers, Avg trip: 1.5 to 7.2 hours, Record/Size: 3600 - 27000 per driver	[49]
Self-collected	CAN-BUS/OBD-II Data	No of Drivers: 20, Age: 20 and 40 years, Avg trip: 10KM, Road type: urban area, which included arterial roads, express lanes, continuous curved roads, a roundabout, and signalized intersections, Time: 9:00–10:30 am and 3:00–4:30 pm Record/size: 87,120 samples	[13], [37]
Self-collected	CAN-BUS/OBD-II Data	No of Drivers: 20, Gender: 14 males, 6 females, Age: 20-54 years Avg trip: 12–17 minutes, 5.7km, Time: 3 pm to 6 pm, 45 days, sunny days, Traffic Type: light traffic, heavy traffic, and other complex scenarios	[14], [86]
Self-collected	CAN-BUS/OBD-II Data	No of Drivers: 45, Gender: 30 males and 15 females Avg trip: 6.5 km in total and takes about 12–15 min per lap Time: sunny days	[23]
Self-collected	CAN-BUS/OBD-II Data	No of Drivers: 8, with 5 different cars, Avg trip: 30 Km, Road type: city streets and highways	[87]
Self-collected	CAN-BUS/OBD-II Data	No of Drivers: 5 Gender: 4 males, 1 female, Avg trip: 17 km), Time: 10–12 A.M.; 2–5 P.M, sunny days	[15]
Self-collected	ECG (Physiological)	No of Drivers: 100 Gender: 64 males and 36 females, Age: 23 - 34 years Affiliation: Chosun University (CU-DB)	[27], [36], [90]
Self-collected	Smartwatch (3-axis accelerometer and 2-axis orientation sensor signals)	Simulator: No of Drivers: 90, Real vehicle: No of Drivers: 20, Avg trip: 1.77 km, Road type: route included five turns, clockwise and counterclockwise	[51]

(to be continued)

Table 3 Detail description of datasets for driver identification which is used in reviewed studies.

(continued)

Dataset	Available features/sources	Detail of Dataset	References
Self-collected	Electronic Stability Control (ESC), Global Positioning System (GPS) and OBD-II	No of Drivers: 57, Avg trip: 153.9 miles, 14 hrs, with each trip: 45 min, Road type: Urban, roundabouts, traffic lights, and streets roads, Time: 10am-12:30pm, 1–5:30pm or 7–10:30pm, london time Link: https://doi.org/10.13140/RG.2.2.14505.49765	[42]
Self-collected	Tri-axial accelerometer	Features: (lateral, longitudinal and vertical acceleration) No of Drivers: 10, Time: 9:00 to 11:00 or 15:00 to 17:00 for 8 days, Record: 5,500,000 each driver, Freq: 100 Hz	[69]
Self-collected	Tri-axial accelerometer	No of Drivers: 25 Trips: 20,025, Each trip: 801 real trips per driver	[16]
Self-collected	IMU (accelerometer, gyroscope, magnetometer	No of Drivers: 10, Trip: 2 hours and 10 minutes, Features: Only accelerometer and gyroscope used	[41]
Self-collected	Audio, video, CAN-bus, gas pedal, brake pedal sensor recordings, and IMU	No of Drivers: 25, Age: 22-57 years, Road type: urban roads, ring roads, high speed roads	[96]
Self-collected	CAN-BUS/OBD-II + GPS Record/size: 292 observations	Link: Home · martinacimitile/Car-Data-Mining Wiki · GitHub	[29], [87]
Self-collected	CAN-BUS/OBD-II + Lidar	No of Drivers: 8, Road type: 4th Ring Road of Beijing	[39]
Self-collected	OBD + GPS	NA	[31]
Self-collected	CAN-BUS/OBD-II Data	NA	[83]
Self-collected	Driving simulator	NA	[91]

can involve the application of random transformations or perturbations to the driving data, such as rotations, translations, or the addition of noise. By implementing data augmentation, the model's generalization capabilities are enhanced, and the risk of overfitting is mitigated. In this regard, four time-series data augmentation techniques—up-sampling, noise injection, data reversal, and random drifting—were employed to enhance model performance^[37]. These techniques were used to enrich the initial training data and improve the performance of an ensemble deep model for driver identification.

Remarkably, the authors achieved an accuracy exceeding 50% even with only one minute of driving data. Their findings highlight the substantial enhancement in model performance that can be attained through augmentation methods, particularly when the training data quantity is limited. In another work^[41], a hybrid model called Generative Adversarial Network-Stacked Generalization Method (GAN-SGM) was introduced for driver identification. The GAN was utilized for data augmentation, while the SGM employing K-nearest neighbor (KNN), MLP, SVM, RF, and LR classifiers was used for classification. The proposed approach demonstrated exceptional performance, achieving 97% accuracy.

4.4 Time window segmentation

Time window segmentation for driver identification involves dividing the continuous time series data, such as sensor readings or driving behavior observations, into smaller, non-overlapping or overlapping segments or windows of fixed duration. This segmentation allows for the analysis and modeling of temporal patterns or behaviors within those specific time intervals. By breaking the data into manageable segments, it becomes easier to capture and analyze the variations and characteristics of driver-specific features over time. Time window segmentation is particularly useful for driver identification as it helps to extract relevant information from the temporal patterns exhibited by drivers during different time intervals. The choice of window size and overlap percentage can impact the accuracy and granularity of the identification process, and it is typically determined through experimentation and optimization based on the specific dataset and identification requirements.

Multiple studies have investigated the application of time window segmentation for driver identification, beginning with a study that achieved 95% accuracy by normalizing OBD-II data and applying a sliding window technique^[29]. Another approach applied the Extra Trees algorithm with sliding window partitioning

Table 4 Pre-processing techniques applied in reviewed studies.

Methods	Strengths	Weaknesses	References
Data Cleaning	Removes noise and outliers, improving data quality. Remove 0-speed data.	May remove useful information along with noise if not carefully applied.	[4], [10], [13], [15], [18], [20], [29], [30], [32], [36], [37], [38], [39], [41], [42], [51], [70], [71], [72], [87], [90]
Feature Scaling	Brings features to a common scale, preventing dominance by a single feature. Helps improve convergence speed and model performance.	Sensitive to outliers if not robustly implemented. Scaling may not be necessary for all algorithms (e.g., tree-based models).	[4], [10], [13], [14], [15], [23], [27], [28], [29], [30], [36], [37], [38], [39], [41], [43], [69], [71], [72], [82], [85], [86], [88]
Data Augmentation	Helps improve model generalization. Reduces the risk of overfitting. Increases the size and diversity of the training dataset.	Augmented data may not accurately represent real-world cases. Augmentation techniques must be carefully chosen to avoid distorting the original data distribution.	[10], [37], [41]
Time Window Segmentation	Captures temporal patterns, extracts relevant features. Identifies driver-specific patterns	Choice of window size and overlap affects results. Determining optimal window size and overlap is challenging.	[4], [10], [15], [16], [18], [20], [23], [29], [32], [37], [39], [40], [41], [69], [71], [72], [81], [82], [85], [86], [88], [89], [96]
Spectrogram	Captures important frequency patterns. Allows for the identification of unique frequency signatures	Spectrogram size and resolution can be sensitive to the chosen windowing parameters. Spectrogram features alone may not capture all driver-specific characteristics.	[16], [36], [90]
Wavelet Transform	Suitable for analyzing non-stationary signals. Can capture both time and frequency information.	Selection of appropriate wavelet function and decomposition levels can be challenging. Computationally more complex compared to other methods.	[15], [41]
Class Balancing	Helps address class imbalance issues by ensuring sufficient representation of minority classes. Prevents bias toward majority classes. Improves model performance for underrepresented classes.	Oversampling techniques may introduce synthetic samples that are not representative of the real data. Undersampling may discard potentially valuable information, leading to an insufficient representation of majority classes.	[21]
Encoding Categorical Variables	Converts categorical variables into numerical representations	May introduce ordinality or non-linear relationships	[43], [42]

of CAN-Bus signals, improving classification accuracy from 96.20% to 99.92% by fine-tuning the window size^[71]. A sliding window method was also used to construct a driving fingerprint map based on selected features, where time window parameters were optimized using deep CNN classification results. Setting the window length and shift to 3 seconds and 1 second, respectively, resulted in 93.5% identification accuracy^[96].

Subsequently, overlapping sliding time windows were explored, with a study achieving 94% accuracy using a 15-minute window and 75% overlap^[18]. A nonnegativity-constrained autoencoder (NCAE) was introduced to identify the optimal sliding window size^[89], while another study used overlapping

segmentation and reported identification accuracies ranging from 97.72% to 98.36%^[88]. Additional analysis of time window length was conducted in another system evaluation^[15]. One study found the best accuracy using a 15-minute window with 75% overlap^[41], whereas another concluded that longer windows improved results, opting for a 30-second window^[32]. A separate investigation determined that a 30-second window yielded an optimal macro F1 score of 89.9%^[13]. Furthermore, an accuracy of up to 99.7% was achieved using only a 10-second driving window and a RF classifier^[42].

In summary, the application of (overlapping) windowing techniques on raw sensor data has proven to significantly enhance the performance of driver

classification systems, particularly those leveraging DL approaches.

4.5 Spectrogram

A spectrogram is a widely used visualization technique in signal processing that allows the analysis of the frequency content of a signal as it evolves over time^[16]. In the context of driver identification, spectrograms can be employed to analyze and represent the frequency characteristics of sensor data collected from a vehicle, such as accelerometer or microphone signals. The spectrogram presents a two-dimensional representation, where the x-axis corresponds to time, the y-axis corresponds to frequency, and the color intensity or shading represents the magnitude or power of the frequency components at each specific time point.

Generating spectrograms from sensor data enables the extraction of patterns and distinctive features related to a driver's behavior, which can then be utilized for identification purposes. These spectrogram-based features are well-suited for input into ML or DL models, offering insights into both the temporal and spectral characteristics of the signals. This facilitates effective analysis and discrimination between different driving behaviors and individuals.

For instance, 1D accelerometer signals were transformed into a 2D format and processed using a discriminative end-to-end DL model to perform driver identification^[16]. Another approach involved converting segmented ECG cycles into 2D spectrograms to improve identification accuracy compared to traditional multidimensional features^[36]. In a follow-up study, the same authors proposed a system that used both ECG and EMG signals, converting them into 2D constant-Q transform (CQT) images. These were then input into a multistream CNN for classification, resulting in a high identification accuracy of 98.9%^[36].

4.6 Class balancing

Class balancing techniques address the issue of class imbalance in the dataset^[21], where certain driver classes may have significantly more samples than others. This may result in biased models that are more accurate on the majority class but perform poorly on minority classes. Balancing techniques such as oversampling, like SMOTE (Synthetic Minority Over-sampling Technique), or undersampling can be used to ensure a more balanced representation of different

driver classes^[21].

4.7 Encoding Categorical Variables

The process of encoding categorical variables entails the conversion of these variables into numerical representations. This can be achieved through various methods such as label encoding or one-hot encoding^[43]. Label encoding gives each category a distinct number value whereas one-hot encoding generates binary columns for each distinct category. These encoding techniques enable categorical variables to be used as input in ML algorithms that require numerical data. It is required for ML algorithms to process them effectively^[42].

4.8 Wavelet transform

Wavelet transform is a signal processing method used in driver identification to analyze the frequency content of signals over different scales^[41]. It decomposes a signal into a series of wavelet coefficients, which represent the signal's energy at various frequencies and time intervals. In driver identification, the wavelet transform is utilized to process sensor data, such as accelerometer or gyroscope signals. This transform helps extract features that effectively capture the distinctive patterns and characteristics specific to individual drivers. The wavelet coefficients provide information about both low-frequency and high-frequency components of the signal, allowing for a detailed analysis of driver behavior.

By analyzing wavelet coefficients and their variations, ML algorithms can effectively classify and identify drivers based on their unique driving patterns. The wavelet transform is particularly advantageous for capturing both transient and stationary features in the signals, making it a powerful tool for driver identification tasks. For example, a system utilizing accelerometer and gyroscope data, combined with the discrete wavelet transform, achieved 97% accuracy in driver identification, surpassing traditional ML methods^[41]. In another study, a wavelet transform-based approach was applied to CAN bus data for driver identification. Using five drivers and an SVM classifier, this method achieved an accuracy rate of approximately 87%^[15].

5 Feature selection methods for driver Identification

Feature selection plays a critical role in driver

identification since it entails carefully selecting a subset of relevant features from a wider set. This process aims to determine the most illuminating and discriminative traits that contribute significantly to the accurate identification of drivers. It plays a vital role by improving model performance, reducing dimensionality, enhancing interpretability, and increasing computational efficiency. By selecting the most informative and discriminative features, feature selection improves classification accuracy and generalization, eliminates redundancy, and reduces overfitting. It enables faster training and inference times, particularly important for real-time applications, while providing insights into the driving behaviors that contribute to driver identification.

The study^[19] emphasized the importance of evaluating the contribution of available features to the classification process in driver identification. It was emphasized that without a comprehensive understanding of how driving behavior influences each feature, it would be challenging to construct reliable and trustworthy models. Various feature selection techniques (Table 5) have been employed for driver

identification in the reviewed studies.

5.1 Univariate Feature Selection

This method involves evaluating each feature individually to determine its connection to the desired variable. To rank the features depending on their relevance, statistical methods such as the chi-squared test, t-test, or ANOVA might be used. Features with high statistical scores are considered more relevant and selected for the model. Experimentation with simple feature selection methods, including univariate feature selection, led to the identification of 40 key features for driver identification^[43].

5.2 Principal Component Analysis (PCA)

PCA is a dimensionality reduction method that converts the original features into a new set of uncorrelated variables called principal components. The greatest amount of data variance is captured by these components. One can successfully minimize the feature space while maintaining the most crucial data by choosing a subset of the top principal components. In a recent study, PCA has been utilized for feature

Table 5 Identified Strength and Weakness of feature selection techniques for driver identification

Technique	Strengths	Weaknesses	References
Univariate Feature Selection	Simple and computationally efficient. Can handle a large number of features. Can uncover individual feature relevance. Statistical significance of features can be determined.	Ignores feature interactions and dependencies. Does not consider the combined effect of features. May overlook useful features if their relevance is dependent on interactions with other features. Assumes features are independent of each other.	[43]
PCA	Reduces dimensionality while retaining most of the information. Allows for visualization of data. Captures major patterns in the data. Reduces computational complexity. Works well with linear models.	May not be suitable for capturing complex non-linear relationships. Interpretability of principal components may be challenging. Does not provide feature rankings or insights into individual feature relevance. Requires standardization of features. May lose some discriminative power if the important features are not the major components.	[4], [14], [16], [23], [28], [43], [52], [68], [85], [86], [97], [98]
Tree-based Feature Importance	Considers feature interactions and non-linear relationships. Handles missing values and outliers. Provides feature rankings based on importance. Works well with categorical and numerical features. Robust to feature scaling. Can handle high-dimensional data.	May prioritize high cardinality categorical variables over continuous variables. Prone to overfitting if the model is excessively complex. Can be biased towards features with high cardinality. Tree-based models can be computationally expensive for large datasets. Feature importance can vary between different tree ensemble methods.	[28], [30], [38], [43], [69], [71], [85]
Correlation-based Feature Selection	Identifies linear relationships between features and the target. Quick and straightforward to compute. Can identify highly correlated features for removal. Provides insights into feature dependencies.	Only captures linear relationships, may miss non-linear dependencies. Ignores feature interactions. May not capture the true relevance of features if there are non-linear relationships. Does not provide feature rankings based on importance. May overlook informative features that are not strongly correlated with the target variable. Requires continuous numerical features for correlation analysis.	[4], [10], [19], [20], [29], [30], [38], [42], [41], [71], [81], [86]
Mutual Information	Considers the statistical dependence between features and the target variable. Captures nonlinear relationships.	May overlook higher-order feature interactions. Sensitive to the discretization or binning of continuous variables.	[96]

pre-processing in driver identification tasks, leading to significant improvements in accuracy. For instance, the study^[4] reported an approximate 20% increase in accuracy.^[68] developed an algorithm that incorporates PCA, specifically focusing on lateral and longitudinal accelerations. Their approach determines driver identification based on the angle between the principal components of test data and those of all training data. Similarly, study^[52] employed PCA to select the most discriminative features associated with drivers from the dataset. Their findings demonstrate that utilizing only relevant features can enhance accuracy while reducing learning time.

The essential procedure for PCA involves a series of steps designed to distill the core elements of the feature space^[97]. The process commences with the computation of the mean $\bar{x}$ of a given feature vector $\{x_1, x_2, x_3, \dots, x_n\}$, articulated as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

Subsequently, the covariance matrix Cov is constructed through

$$Cov = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T \quad (2)$$

where x_i denotes the mean-adjusted data vector. The eigenvalues and eigenvectors are then extracted from Cov via the relation

$$Cov \times e_k = \lambda_k \times e_k \quad \text{for } k = 0, 1, \dots, n-1 \quad (3)$$

where e_k signifies the k th eigenvector and λ_k the corresponding eigenvalue. To curtail the dimensionality of the feature vectors, one selects the eigenvectors associated with the most substantial eigenvalues. The matrix of selected eigenvectors is represented as

$$M_{fv} = (e_1, e_2, e_3, \dots, e_p) \quad (4)$$

where p corresponds to the number of dimensions retained in the PCA matrix. The transformed feature vector is then given by

$$N_{fv} = M_{fv}^T \times T \quad (5)$$

with T being the transpose of the mean-adjusted original input vector.

In a recent study,^[98] successfully applied PCA to condense the dimensionality of audio scripts by linearly projecting the frequency components into a lower-dimensional space, retaining only 80 of the

original 256 dimensions post-PCA application.

5.3 Tree-based Feature Importance

Ensemble methods, such as RF and Gradient Boosting, compute feature importance by evaluating how each feature influences the model's predictive capability. This process ranks features based on importance scores, facilitating the selection of the most relevant ones. In the context of driver identification, Extra Trees were employed to achieve a recognition accuracy of 88%^[71]. A separate feature selection approach involved identifying key features using techniques such as Coefficient analysis, tree-based methods, and PCA^[38]. The findings revealed that tree-based methods, particularly the XGBoost algorithm, delivered the best results with 75% accuracy and reduced training time. Extra Trees also demonstrated the highest performance in another study, achieving 98.47% accuracy using the top 15 features extracted from CAN-bus data^[85]. Similarly, tree-based classifiers—specifically RF—were found to be the most effective compared to other models for the dataset in a separate study^[28]

5.4 Correlation Analysis

This technique involves analyzing the correlation between each feature and the target variable. High correlation features are likely to be more informative for driver identification and can be selected for the model^[29, 86]. Additionally, highly correlated features with each other can be removed to avoid redundancy.

5.5 Mutual Information

Mutual information measures the dependency between variables and can be used to quantify the relationship between each attribute and the target variable. Features with higher mutual information scores are considered more relevant and are prioritized for selection. The Conditional Mutual Information Maximization (CMIM) algorithm, which evaluates both the relevance of features to the target class and their intercorrelation, was used to perform feature selection for constructing a driving fingerprint map in driver identification^[96]. According to the CMIM results, the most important variables for accurately identifying drivers were the distance to the lane, acceleration along the X-axis, and yaw rate.

6 Machine Learning (ML) methods for driver Identification

ML is a field within artificial intelligence that

concentrates on the creation of algorithms and models capable of enabling computers to learn and make predictions or decisions autonomously, without the need for explicit programming instructions. The process involves training these models using vast quantities of data to uncover patterns, relationships, and underlying structures. Once the models are trained, they possess the ability to generalize from the training data and apply their learned knowledge to make predictions or classifications on new, unseen data.

Driver identification has been addressed using different ML algorithms (Table 6), such as the GMM, SVM, DT, and RF. We have included the most recently applied ML algorithm in this study.

6.1 GMM

GMM is a probabilistic model employed to analyze and represent intricate data distributions. The GMM assumes that the observed data is generated from a combination of Gaussian distributions, where each Gaussian component represents a distinct subpopulation within the data. A GMM is conceptualized as a multivariate probability distribution approach^[99]. It is adept at modeling arbitrary distributions and has become a prevalent method in Speaker Identification systems^[100]. The GMM for a feature vector distribution pertaining to a speaker s is represented as a weighted linear combination of M unimodal Gaussian densities $b_{s_i}(\mathbf{x})$, parameterized by mean vectors $\boldsymbol{\mu}_{s_i}$ and a covariance matrix $\boldsymbol{\Sigma}_{s_i}$. These parameters collectively form a speaker's model, denoted as $s_i, \boldsymbol{\mu}_{s_i}, \boldsymbol{\Sigma}_{s_i}\}_{i=1}^M$. The mixture weights p_{s_i} comply with the stochastic constraint

$\sum_{i=1}^M p_{s_i} = 1$. For a feature vector $\mathbf{x}$, the density mixture for speakers is computed as:

$$p(\mathbf{x}|\lambda_s) = \sum_{i=1}^M p_{s_i} b_{s_i}(\mathbf{x}) \quad (6)$$

where each Gaussian component $b_{s_i}(\mathbf{x})$ is defined as:

$$b_{s_i}(\mathbf{x}) = \frac{1}{(2\pi)^{D/2} |\boldsymbol{\Sigma}_{s_i}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_{s_i})^T (\boldsymbol{\Sigma}_{s_i})^{-1} (\mathbf{x} - \boldsymbol{\mu}_{s_i})\right) \quad (7)$$

and D represents the dimension of the feature vector. Given a set of feature vectors $\mathbf{X} = \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T$ corresponding to a driving behavior signal spanning T frames, the log-probability for a specific driver model s is computed as:

$$L_s(\mathbf{X}) = \log p(\mathbf{X}|\lambda_s) = \sum_{t=1}^T \log p(\mathbf{x}_t|\lambda_s) \quad (8)$$

In the context of Driver Identification (DI), the value of $L_s(\mathbf{X})$ is determined for each enrolled driver model λ_s , and the model yielding the maximum likelihood is identified as the corresponding driver. During the training phase, the feature vectors are refined using the Expectation-Maximization (EM) algorithm^[101, 102], a method involving iterative updates to each parameter in λ , thereby enhancing the log-probability at each iteration.

GMM is utilized in driver identification by modeling the driving patterns of different individuals as separate Gaussian components. GMM can effectively differentiate between different drivers by capturing their distinct driving behaviors and patterns by estimating the parameters of these components, such as

Table 6 Identified strengths and weaknesses of each ML algorithm used for driver identification in reviewed studies

Method	Strengths	Weaknesses	References
GMM	Can model complex data distributions. Can handle overlapping or multimodal clusters. Probabilistic framework allows uncertainty modeling. Effective in high-dimensional feature spaces.	Can be computationally expensive for large datasets. Initialization can impact the quality of results. Can struggle with high-dimensional data.	[39], [51]
SVM	Robust against overfitting. Can handle non-linear classification boundaries. Memory efficient during inference.	Computationally expensive during training. Requires careful hyperparameter selection. Sensitivity to noisy or irrelevant features. Lacks probabilistic output by default.	[15], [32], [49], [51]
DT	Easy to understand and interpret. Can handle both numerical and categorical features. Non-parametric and distribution-agnostic. Provides feature importance.	Prone to overfitting, especially with deep trees. Sensitive to small variations in the training data. Lack of robustness when dealing with noisy or irrelevant features. Limited generalization ability.	[31], [52]
RF	Effective in handling complex interactions. Robust against overfitting. Provides feature importance measures.	Computationally expensive during training. Lack of interpretability for individual trees. Requires careful hyperparameter tuning. May struggle to generalize well with unrepresentative training data.	[19], [20], [21], [38], [42], [49], [69], [70], [71]

means and covariances. This approach facilitates the development of driver identification systems capable of identifying and authenticating individuals based on their unique driving characteristics.

Some researchers have employed GMM^[51, 39] for driver identification. This algorithm has shown promise in capturing drivers' complex patterns and behaviors based on various input data sources. By modeling the data as a mixture of Gaussian distributions, GMMs can effectively cluster and identify different driver profiles. Studies have leveraged GMMs to analyze features such as driving behavior, vehicle telemetry, or sensor data, providing valuable insights into driver identification. The utilization of GMM for driver identification demonstrates its effectiveness and adaptability in this specific domain. In their study^[51] present a driver identification system that specifically targets the analysis of drivers' maneuvers, with a specific emphasis on right and left turns. The authors employ GMM-based approaches and report an error rate of approximately 18% for driver identification. Study^[39] presented a driver identification approach that considers the heterogeneity within drivers and between different drivers in car-following sequences. They utilize a GMM to model these variations. By conducting an experiment with eight drivers, collecting car-following samples of 15 seconds per sample and 10 samples per driver, the proposed method demonstrates a driver identification accuracy of 82.3%.

6.2 Support Vector Machine (SVM)

SVM is a widely employed ML algorithm utilized for classification and regression tasks^[103, 104]. It operates by identifying an optimal hyperplane that efficiently separates data points from different classes. This is achieved by maximizing the margin, which represents the distance between the hyperplane and the closest data points. SVMs are powerful in dealing with both linearly separable and non-linearly separable data by employing kernel functions. These kernel functions enable SVMs to transform the input space, effectively separating complex, non-linear patterns in the data. They are renowned for their capability to handle high-dimensional feature spaces and their resistance to overfitting.

SVM hinge upon the construction of an optimal hyperplane that maximizes the margin between two distinct classes, as originally introduced and further developed in related methodologies^[105, 106]. For tasks

involving multinomial classification, SVM adopts a structural risk minimization framework^[107], which can be formalized as:

$$\min_{\mathbf{w}, \xi_k} J_1(\mathbf{w}, \xi_k) = \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^N \xi_k \quad (9)$$

The linear SVM methodology^[108], entails the projection of an input feature vector $\mathbf{x}$ into a functional space, $f(\mathbf{x})$, delineated as:

$$f(\mathbf{x}) = \text{sign} \left[\sum_{i=1}^N \alpha_i k(\mathbf{x}; \mathbf{x}_i) + \beta \right] \quad (10)$$

In these equations, $\mathbf{x}_i$ represent the support vectors, β is the bias term, α_i are the Lagrange multipliers or weights, and N denotes the total number of support vectors.

SVM have been extensively utilized as a primary algorithm for classification tasks due to their robustness and effectiveness in handling high-dimensional data. SVM have been utilized by researchers to conduct driver identification studies^[109, 49, 79, 15]. An event-driven framework is introduced for driver identification and impostor detection, utilizing features extracted from GPS data^[49]. The framework employs SVM as the supervised classification method. To achieve high accuracy in scenarios with many drivers, the researchers propose a modified radial basis function (RBF) kernel. The empirical evaluation demonstrates promising results, with an average recall of 94% for ten drivers and 90% for 24 drivers. Furthermore, the framework achieves impostor driver detection with over 90% accuracy when dealing with three or more drivers. CAN bus data was utilized with an SVM classifier to perform driver classification, achieving up to 95.07% accuracy for binary classification and 89.06% for multiclass classification^[15].

6.3 Decision Tree (DT)

A DT is a widely adopted ML algorithm that employs a tree-like model for making decisions or predictions. It organizes data into a hierarchical structure consisting of nodes and branches. Each internal node within the tree represents a feature or attribute, whereas each branch corresponds to a decision made based on that particular attribute. The tree structure is constructed by iteratively dividing the data using various features in a recursive manner, aiming to create subsets that are as pure as possible in terms of the target variable. DTs can

handle both classification and regression tasks and are interpretable, allowing users to understand the decision-making process. They are known for their versatility, ease of implementation, and capability to handle both categorical and numerical data. DT algorithm was utilized to assess the effectiveness of the proposed method in driver identification^[52]. The dataset^[17] used consists of 51 distinct features extracted from 10 different drivers, and the evaluation demonstrates a precision and recall of 99%.

6.4 Random Forest (RF)

RF is a widely used ML algorithm that constructs an ensemble of DTs^[110]. By combining the predictions of multiple individual DTs, it enhances the accuracy and robustness of the overall predictions^[111]. Each DT is trained on a random subset of the training data, and during the prediction phase, the final result is determined by aggregating the predictions of all the trees. RF is known for its effectiveness in handling high-dimensional data, as well as capturing intricate relationships within the data. Additionally, it serves as a useful tool for mitigating overfitting, a common challenge in ML. It is commonly employed algorithm for both classification and regression tasks, and its versatility and performance have made it widely adopted in various domains.

7 Deep Learning (DL) methods for driver identification

DL is a branch of ML that utilizes artificial neural networks to learn patterns and make predictions from data^[112]. Inspired by the human brain, these neural networks are designed to mimic the structure and functionality of the brain's interconnected neurons, excelling at detecting intricate patterns within the data^[113]. DL has been successfully employed to address various challenges, including object recognition, speech recognition, face recognition, sentiment analysis, and numerous classification tasks^[114]. In recent years, DL approaches have shown promising performance across different domains, including network communication failure recovery^[115], Cyber security^[116, 117, 118], intelligent transportation systems and smartcity^[119, 120, 81]. Different types of DNNs (Table 7) include CNNs, RNNs, and deep autoencoders, commonly employed for driver identification tasks. Deep autoencoders, specifically, are neural networks designed to reconstruct the input at the output, primarily utilized for reducing the dimensionality of data.

7.1 Deep Neural Network (DNN)

DNN is an artificial neural network that comprises multiple hidden layers positioned between the input

Table 7 Deep Learning Techniques for Driver Identification

Method	Strengths	Weaknesses	References
DNN	Capable of learning complex non-linear relationships between input features and driver identification. Versatile and widely applicable to various domains. Handles high-dimensional data effectively.	Requires careful hyperparameter tuning and architecture design. Susceptible to overfitting, especially with limited training data. Prone to getting stuck in local optima during training. Computational complexity increases with the number of layers and neurons.	[18], [28], [29], [72], [121]
CNN	Effective in capturing spatial features. Robust to input variations. Hierarchical feature extraction. Parameter sharing and regularization.	Limited temporal modeling. Lack of interpretability. Computationally expensive.	[13], [24], [36], [85], [89], [96], [122, 123, 124, 125]
LSTM	Effective in capturing long-term dependencies. Mitigates the vanishing gradient problem. Handles sequential data effectively. Efficiently learns low-dimensional representations of data. Reduces dimensionality and extracts important features.	Computationally expensive. Difficulty in parallelization. May suffer from overfitting if not properly regularized.	[27], [40], [43], [83], [126]
DAE	Preprocesses high-dimensional sensor data effectively.	Unsupervised learning requires labeled data for driver identification. Reconstruction loss may not directly correlate with driver identification accuracy. Sensitivity to hyperparameter tuning and network architecture.	[89], [127]
Hybrid Model	Captures both spatial and temporal features effectively. Hierarchical feature extraction. Memory of past information. Variable-length inputs. Handles sequential and spatial data.	Computationally expensive due to the combination of CNN and RNN. Requires careful architecture design and hyperparameter tuning. Potential difficulty in interpretability of combined CNN and RNN components.	[4], [10], [16], [30], [32], [37], [41], [88], [86]

and output layers, as depicted in Fig. 6. It is called “deep” because of its depth, meaning it has a large number of layers compared to traditional neural networks^[121]. Each layer in a DNN contains multiple interconnected neurons that perform computations on the input data. The architecture of a DNN typically adopts a feedforward structure, where information propagates in a unidirectional manner, starting from the input layer and passing through the hidden layers before reaching the output layer. In each hidden layer, every neuron receives inputs from the preceding layer, calculates a weighted sum of these inputs, applies an activation function to the sum, and then transmits the result to the next layer in the network. This iterative process continues for each layer until the output layer generates the final prediction or classification based on the transformed inputs from the preceding layers.

Training a DNN consists of two primary stages: forward propagation and backpropagation. During the forward propagation phase, the input data is fed into the network, and predictions are generated as the data flows through the network. The disparity between the predicted outputs and the actual outputs is assessed by employing a loss function. Subsequently, backpropagation is employed to compute the gradients of the loss function in relation to the weights and biases of the network. The gradients calculated in the previous step are utilized to update the weights and biases of the

network through optimization algorithms such as gradient descent.

$$\Delta\omega_{ij}(t+1) = \Delta\omega_{ij}(t) + \eta\sigma C/(\sigma\omega\partial\omega_{ij}) \quad (11)$$

In the given equation, the learning rate is denoted by η , the weight is represented by ω_{ij} , and the cost function associated with the weights is denoted by C .

Driver identification using DNN has shown promising results across several studies. One study identified drivers within a group of 10 using a DNN model and achieved an accuracy of 95%^[29]. Feature reduction techniques, specifically PCA and RF, were combined with a DNN model to improve performance, resulting in 97.05% accuracy for the RF-DNN model and 95.55% for the PCA-DNN model^[28]. Another proposed DNN-based approach achieved an 81% accuracy rate with a testing duration of just 13 seconds using data from two smartphone sensors—an accelerometer and GPS^[72]. A separate study reported an impressive 99% accuracy by applying a DNN to histograms derived from acceleration data captured by the accelerometer sensor^[18].

7.2 CNN

CNN^[122] is a specialized DL architecture designed specifically to process and analyze structured grid-like data, such as images or sequential data^[123]. While CNNs are commonly utilized for computer vision applications, their applicability extends beyond that

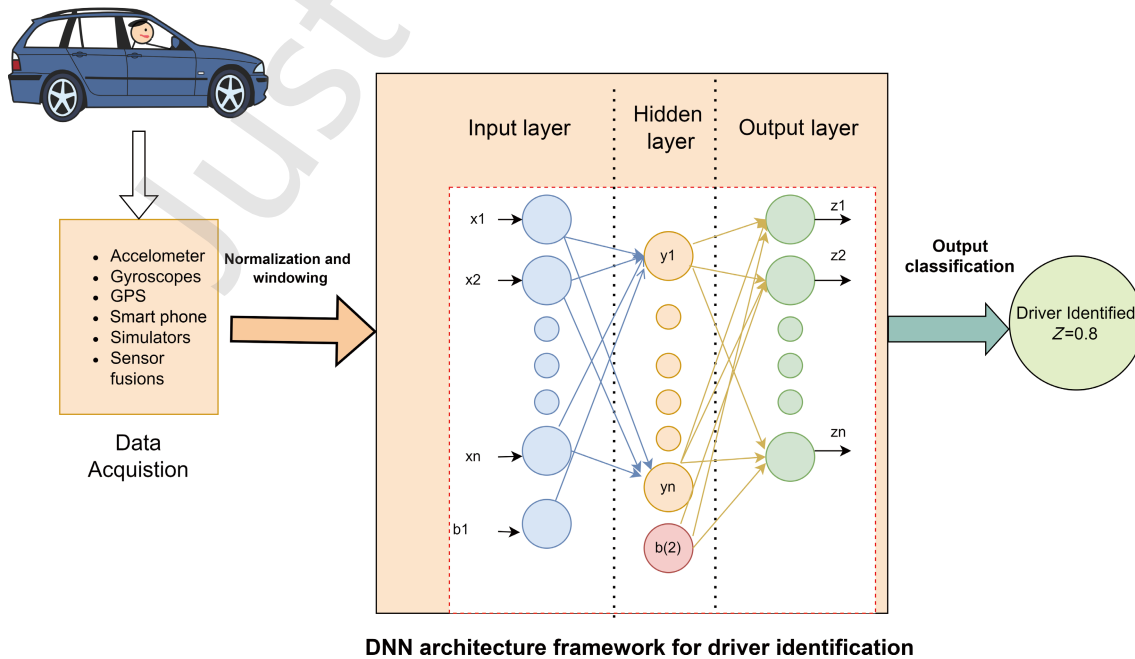


Fig. 6 General deep neural network architecture framework for driver identification system.

domain. CNNs can be employed in various fields, including natural language processing and speech recognition^[24]. The fundamental concept underlying CNNs involves utilizing convolutions, a mathematical operation that enables the integration of input data with learnable filters or kernels^[123]. This process allows for the extraction of pertinent features that are essential for the network's learning and decision-making. The process of convolution entails sliding the filters across the input data, performing element-wise multiplications, and subsequently summing the results to generate a feature map. This process is repeated for different filters, creating multiple feature maps that capture various patterns and spatial hierarchies within the input.

A CNN is commonly structured with multiple layers organized hierarchically^[124]. The fundamental components of a CNN comprise convolutional layers, activation functions, pooling layers, and fully connected layers. Convolutional layers play a crucial role in a CNN as they apply filters to the input data, extracting relevant features. Activation functions are utilized to introduce non-linearity to the network, enabling it to capture complex relationships within the data. Pooling layers perform downsampling of the feature maps, decreasing their spatial dimensions while preserving essential features. Fully connected layers establish connections between every neuron in a given layer and every neuron in the subsequent layer, facilitating the representation of high-level features and final classification. Fig. 7 depicts the CNN structure.

The initial CNN architecture was introduced in 1998, primarily for recognizing handwritten digit images^[125]. In the context of driver identification, the main challenge lies in extracting and capturing the distinctive characteristics of each driver's behavior. CNNs offer a promising solution, as they can automatically learn and extract relevant features from driving data, enabling effective driver classification. One study focused on using a CNN model to

distinguish between drivers, employing an architecture that included a convolutional layer, max-pooling, and a fully connected layer^[14]. By extracting over 20 features from CAN-bus signals—including accelerator and brake pedal usage, as well as engine speed—the model achieved an accuracy exceeding 95% across two different vehicle types. Another study proposed a real-time CNN model, the three-stream convolution network (TS-CNN), for driver identity verification^[24]. Based on the lightweight SqueezeNet architecture, their model effectively captured features from low-quality facial images, achieving an average verification accuracy of 87.44%.

A driver identification system employing a one-dimensional CNN (1D-CNN) architecture was developed to enhance classification performance^[13]. The architecture includes two convolutional-pooling layers, a fully connected layer, and a SoftMax layer. This approach outperformed SVM, MLP, and LSTM models in terms of macro F1 score, achieving an impressive 99.1% accuracy in identifying 20 drivers. The 1D-CNN also demonstrated stable performance and robustness, making it a strong candidate for driver identification tasks. An improved version of CNN, known as the residual convolution network (RCN), was introduced for real-time driver identification^[85]. The RCN architecture consists of four conventional CNN layers, followed by max-pooling and dropout layers. This method achieved high performance, with a training time of under two hours and an accuracy of 99.3%. Comparative analysis showed that the RCN approach outperformed traditional ML algorithms and offered slightly better performance than both RNN and CNN-RNN hybrid models.

A CNN-based driver identification system using 2D spectrogram multidimensional features was proposed to process segmented ECG cycles^[90]. This method optimized time-frequency resolution, resized the 2D image output, and performed driver identification using a single CNN. Additionally, a multi-stream CNN

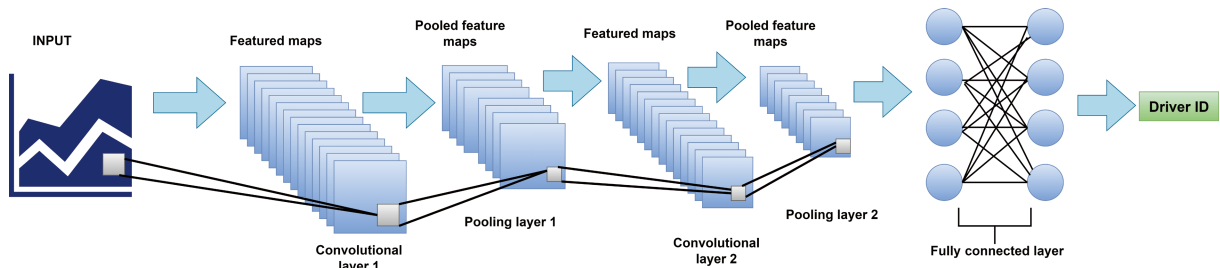


Fig. 7 General convolution neural network architecture for driver identification system.

model leveraging both ECG and EMG signals was proposed, achieving an accuracy of 98.9% in driver recognition^[36]. Another study employed a CNN with conditional mutual information maximization to generate driving fingerprint heat maps, which effectively captured distinctive driving patterns^[96]. This approach achieved a notable driver identification accuracy of 93.5

7.3 RNN

RNN is a specialized form of artificial neural network created to handle sequential data or data with temporal dependencies^[81]. In contrast to feedforward neural networks, RNNs incorporate internal loops within their architecture, enabling them to retain information and maintain temporal dependencies over time. This enables them to capture and analyze patterns in sequential data by incorporating information from both the current input and previous time steps. At every time step, an RNN receives an input, generates an output, and updates its hidden state. The hidden state acts as the memory of the network, enabling it to store information from previous inputs. This hidden state is fed back into the network at the next time step, making RNNs capable of modeling sequences of arbitrary length. Fig. 8. shows the architecture of RNN. The key component of an RNN is the recurrent connection, which enables the flow of information from one time step to the next. This connection allows the network to learn long-term dependencies and contextual information from the input sequence. By considering the context and history of the data, RNNs have shown great efficacy in tasks like language modeling, speech recognition, machine translation, and time series prediction.

7.3.1 Long Short-Term Memory (LSTM)

A widely used form of RNN is the LSTM network, which tackles the issue of vanishing gradients in conventional RNNs through the use of memory cells and gates that control the flow of information^[126]. These mechanisms enable LSTMs to selectively retain and retrieve information from previous steps, making them highly effective in capturing long-term dependencies in sequential data.

The LSTM network has been effectively utilized to capture the driving behavior patterns of individuals, enabling the network to retain memory and learn time-dependent characteristics of driving behavior^[23]. This method allows for the learning of inter-feature relationships and the extraction of meaningful insights from temporal driving data. Building on this approach, an optimized DL model leveraging LSTM networks was introduced for driver identification^[43]. The architecture included an input layer, two LSTM layers with hidden layer dropout regularization, and a dense output layer with a softmax activation function. To further enhance performance, hyperparameter tuning was conducted using the Crow Search Algorithm (CSA), a nature-inspired optimization technique. This optimization process led to an impressive driver identification accuracy of 99

Research conducted by LSTM networks for driver identification utilized by^[27]. By analyzing the normalized ECG signal, an impressive accuracy of 96.5% was achieved by them^[27]. In a separate study on driver identification, an LSTM network architecture consisting of four hidden layers was utilized in the study^[83], comprising two LSTM layers and two dense layers. The LSTM layers were comprised of 64 hidden units, while the first dense layers contained 32 hidden

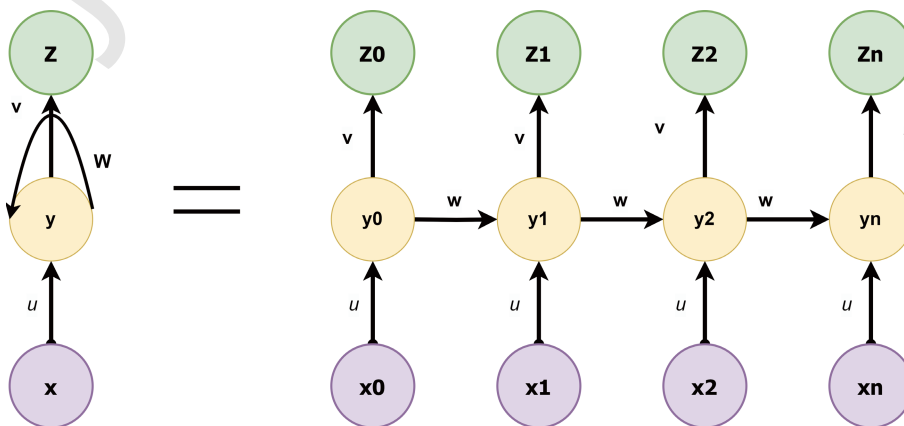


Fig. 8 RNN architecture.

units, and the second dense layers comprised 1 hidden unit. By leveraging real-world driving data, the researchers achieved an impressive overall accuracy of $81\% \pm 7.5\%$. Furthermore, LSTM networks were utilized by [110], with results indicating that bidirectional LSTM networks outperformed forward LSTM networks, demonstrating an average accuracy improvement of 7% in driver identification performance. This highlights the effectiveness of bidirectional LSTM architectures in capturing temporal dependencies and enhancing the accuracy of driver identification tasks.

7.4 Deep Autoencoder (DAE)

A DAE is a sophisticated neural network architecture that incorporates multiple hidden layers between the encoder and decoder components of an autoencoder^[127]. It extends the traditional autoencoder by allowing the network to learn hierarchical representations of the input data. The encoder maps the input to a lower-dimensional latent space through successive hidden layers, capturing different levels of abstraction. The decoder then reconstructs the original input from the latent space, gradually increasing the dimensionality. The DAE learns to reconstruct the input by minimizing a reconstruction loss, enabling it to capture complex patterns and extract meaningful features.

Several approaches aim to extract hidden features from driving data for the specific objective of driver identification. The use of a deep nonnegativity constrained autoencoder for feature extraction purposes was introduced^[89]. Their proposed method included window size optimization for data segmenting and extracting hidden features through a layer-by-layer greedy approach. The extracted features were passed through a SoftMax classifier, which generated probabilities for each driver, enabling driver identification. While this method requires substantial computational resources for data calculation and training, it exhibits high accuracy in performance. Overcoming the challenge of graphics processing unit (GPU) usage could potentially broaden the applicability of these DL algorithms to diverse types of data.

7.5 Hybrid Model

To maximize the advantages of different base models, researchers have explored using hybrid models for

multiple applications. Study^[88] proposed a hybrid DL architecture based on an attention-based combination of CNN, Gated Recurrent Unit (GRU), and LSTM. This approach allows the models to be trained using raw data without manual feature engineering. The neural network layers extract the hidden temporal dependencies within the data. The advantage of this approach is that it leverages CNNs to automatically extract relevant features without the need for explicit feature selection. However, a potential drawback is that the models require many training epochs.

A hybrid model called GAN-SGM is presented for driver identification in the study conducted by^[41]. The GAN is utilized for data augmentation, while the SGM employing KNN, MLP, SVM, RF, and LR is used for classification. The hybrid GAN-SGM approach demonstrates exceptional performance with an accuracy of 97%, precision of 98%, recall of 97%, and F1-measure of 97%, surpassing conventional ML techniques. Researchers [16] have directed their attention towards the utilization of DL techniques for driver identification and verification. They have introduced a hybrid model architecture called CNN-RNN for driver identification. This architecture incorporates ResNet-50 as the initial component, followed by two Stacked Gated Recurrent Units (SGRUs). A Dense layer and a final SoftMax layer are included in the model as well. For driver verification, the authors have proposed the utilization of Siamese Neural Networks and Triplet Loss Training. In their implementation, ResNet-50 is employed as the CNN feature extractor, and GRUs are used for modeling the recurrent dynamics. Driver verification received an F1 score of 74.09%, while driver identification produced top-1 and top-5 accuracy results of 71.89% and 92.02%, respectively.

Given that the sensors used to collect data for driver identification are not entirely immune to malfunctions or failures^[10], it is common to encounter anomalies within the collected sensor data. Therefore, it is crucial to have a well-defined strategy for detecting and correcting these anomalies to ensure stable and reliable results. Study^[10] investigated the importance of detecting and correcting data anomalies in driver identification schemes by integrating CNN and LSTM. They evaluated the performance of their method by comparing the results with and without data anomaly detection techniques. The findings demonstrated that

incorporating anomaly correction, even with varying anomaly rates, led to improved results. However, a notable limitation is the requirement for a substantial amount of driving data (typically training samples) to achieve such high-performance outcomes using these approaches.

Obtaining accurately labeled data from multiple drivers, including both vehicle owners and thieves, for supervised learning is practically impossible. Therefore, anomaly detection has emerged as a preferred approach for vehicle theft detection. In their work^[4], propose a redesigned GAN model called Convolutional Long short-term GAN (CLGAN) for theft detection. The CLGAN model combines LSTM and a CNN and only requires accessible data from the owner of the vehicle. The average accuracy achieved by the proposed CLGAN model is an impressive 98.5%, with the highest accuracy of 92.1% observed across various road types. However, it should be noted that the CLGAN model incurs increased time and memory costs. To address the need for immediate driver behavioral analysis in real time, the focus should be on enhancing the system's ability to identify the thief as early as possible.

Various approaches have been explored to address the identification of illegal or unauthorized drivers. A scheme for automobile driver fingerprinting using a combined model of CNN and Support Vector Domain Description (SVDD) was proposed^[86]. Their approach incorporates an unsupervised anomaly detection method, SVDD, to identify imposters. The authors recommended employing the output of a CNN model as input for the anomaly detection procedure. The findings indicated an accuracy exceeding 95% for two different vehicle categories^[86]. A methodology called the Siamese Temporal Convolutional Network was proposed^[81], which combines the concept of Siamese networks with a 1D temporal convolution architecture for driver verification with a limited duration of driving data. They extended the driver verification approach to include the detection of impostors in an open-set scenario, eliminating the requirement for model re-training. However, a limitation of their method is that the decrease in accuracy as the number of drivers increases^[81].

A multi-task learning network (MTL) was introduced to address the tasks of detecting illegal drivers, identifying legal drivers, and evaluating driver behavior. Their MTL network consisted of LSTM,

SVDD, and feedforward neural network components. The LSTM module was used to identify driving behavior traits, and its output was used as input for the SVDD and feedforward neural networks to detect unlawful drivers and identify legitimate drivers. The outcomes showed that the proposed MTL network achieved high accuracy, broad application, and strong resilience across the three tasks, enabling parallel learning and lowering time and space expenses^[23]. A design of a DL architecture, particularly a stacked encoder model for driver identification, was proposed. This architecture utilizes dilated causal convolutions, deep residual blocks, and triplet loss to effectively extract driver embeddings from data without supervised learning. These embeddings are then used to differentiate among drivers belonging to a specific group. The proposed solution achieves an average accuracy of 94.7% for two-way identification and 96% for three-way identification. Notably, their approach outperforms supervised methods when utilized with data that has limited labeling^[32].

A deep meta-learning approach called "MetaARNet" was proposed to solve the issues with deep model adaptation with varying numbers of drivers (output classes) when there are just a small number of driving samples available^[82]. Their method, referred to as MetaFSDI (Meta Few-Shot Driver Identification), leverages deep meta-learning to enable fast adaptation in such scenarios. The authors adopt a meta-learning framework due to its advantage in reusing features for domains with similar tasks. To ease the development of generalised driving style representations, they introduce an autoencoder regularised convolutional neural feature extractor network (ARNet). This network aids in capturing essential characteristics of driving behaviors. To evaluate the performance of their approach, experiments are conducted on 3-way and 5-way problems using 5-shot and 1-shot approaches, respectively, using a publicly available driving dataset. The results demonstrate that their method outperforms traditional meta-learning algorithms, ensuring better adaptation performance in few-shot driver identification scenarios.

An ensemble DL framework^[37], combines a modified M1-D CNN and bidirectional long short-term memory (BLSTM) to improve robustness, prevent overfitting, and enhance generalization. After employing four different data augmentation techniques, the proposed model attained an accuracy exceeding

50% when limited to one minute of driving data. In another study^[30], developed an ensemble approach known as stacked generalization, which combines multiple models. They utilized stacked generalization with various ML methods, including KNN, SVM, DT, and RF, using Floating Car Data (FCD) collected from smartphones. The system achieved an identification accuracy of 88%. However, it should be noted that their approach was not scalable for all drivers, suggesting potential limitations in its application.

8 Large Language Model for driver identification

LLMs like OpenAI's Chat-GPT are revolutionizing Natural Language Processing (NLP). These models, based on advanced DL architectures such as Transformers, excel in understanding and generating language, rivaling human-like conversation and content creation abilities. Their sophistication is rooted in processing vast datasets, where larger data inputs enhance their output's nuance and accuracy. A key

aspect of LLM development is the labor-intensive text labeling process. Human annotators play a vital role in this phase, organizing and tagging enormous data volumes to provide structured context. This meticulous task significantly enhances LLMs' language processing capabilities.

Recently, LLMs have made a significant leap, gaining the ability to autonomously generate labeled datasets. This breakthrough has transformed their development, shifting away from the previously manual-intensive labeling process. This automation not only streamlines but also accelerates LLM development, marking a pivotal change in their evolution. Our previously paper^[128] has already given overview to reader for better understanding the architecture and performance comparison of different LLMs such ChatGPT, Google Bert etc.

8.1 Framework for driver identification using LLM

Applying LLM in driver identification involves

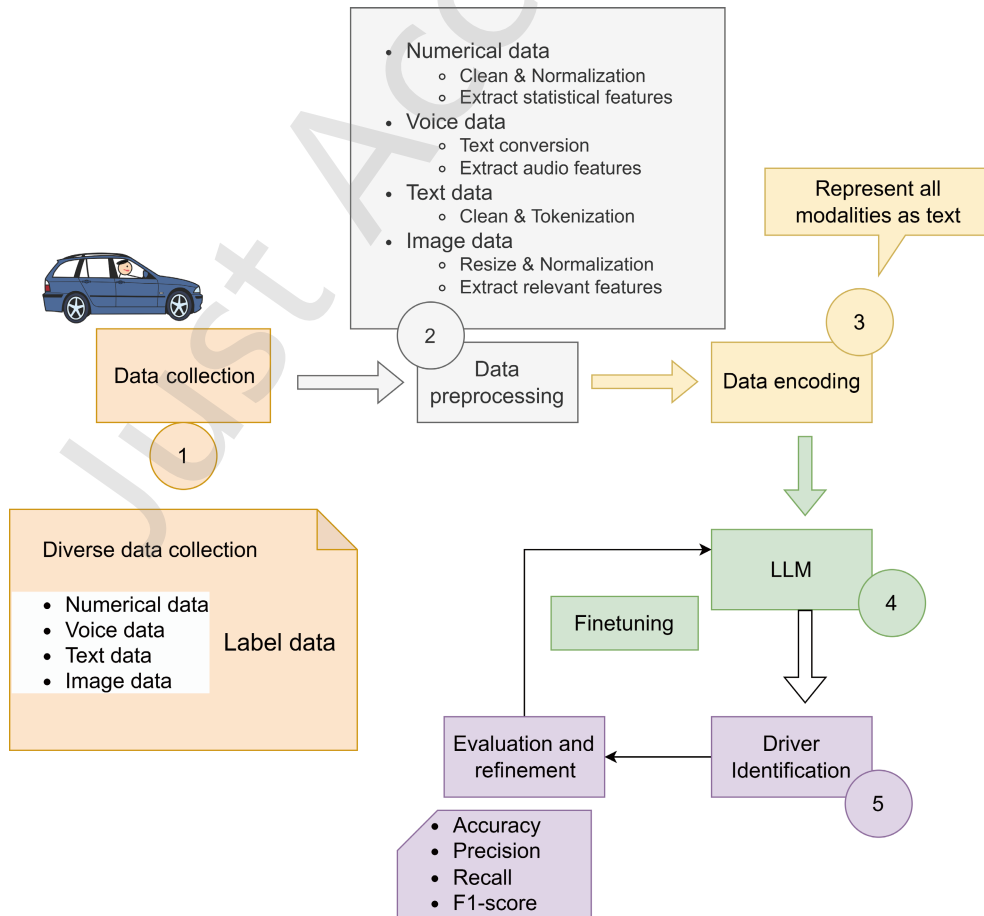


Fig. 9 Advanced driver identification through LLM with different nature input data.

integrating the model's NLP capabilities with driver-related data to enhance accuracy and personalization. This section explains the framework components used for driver identification.

8.1.1 Data Collection and Preprocessing

Gather a diverse and comprehensive dataset of driving-related information, including driving behavior data, vehicle telemetry, sensor readings, driver profiles, and any other relevant data points. Ensure that the data is labeled with driver identification information.

- **Gather multimodal data:**

- Numerical data: Collecting numerical data is an important aspect of understanding driving patterns and behaviors. The data collected may include GPS coordinates, vehicle speed, acceleration, and various sensor readings. Such quantitative insights can be used to identify patterns and trends in driver behavior, which can be essential for improving driving safety and efficiency. By analyzing this data, we can develop better strategies for fleet management, route optimization, and fuel consumption. It can also help in identifying areas where driver training and coaching may be required. Overall, collecting and analyzing numerical data can provide valuable insights into various aspects of driving, which can lead to a safer and more efficient driving experience.

- Voice data: One crucial type of data that can be collected is voice data, which includes audio recordings of the driver's interactions such as commands, conversations, or phone calls made while driving. This type of data can offer valuable insights into the driver's communication style, verbal cues, and behavior behind the wheel. By analyzing this data, it is possible to gain a deeper understanding of how drivers interact with their vehicles and the environment around them. This information can be used to help improve driver safety, optimize vehicle performance, and develop more effective communication tools for drivers.

- Text data: Gathering and analyzing various sources of data such as driving logs, navigation history, social media posts, and text messages can provide valuable insights into the habits and preferences of the driver. For instance, compiling driving logs can help identify frequent destinations, routes, and driving patterns. Navigation history can reveal the places the driver often visits, while social media posts and text messages can offer additional context about their interests, activities, and social connections. By analyzing this

text data, it is possible to gain a better understanding of the driver's behavior, preferences, and tendencies, which can be useful for various purposes such as improving safety, personalizing services, or detecting anomalies.

- Image data: One of the key components of ADAS and autonomous vehicle technology is the collection and analysis of visual data. This includes gathering various types of images such as facial images, in-vehicle camera captures, and road scene images. These visual data inputs can be used to recognize and analyze the driver's physical features, as well as their interactions within the vehicle. By gathering this information, ADAS and autonomous vehicle systems can make more informed decisions and take appropriate actions to ensure the safety of the driver and passengers.

Accurately labeling each data point with the corresponding driver ID is a crucial step in building a robust driver behavior model. When we label each data point with its respective driver ID, we enable the model to learn and identify patterns that are unique to each driver. This process is essential because it helps the model to differentiate between different driving styles and behaviors. In the absence of accurate labeling, the model may not be able to associate the patterns it identifies with specific drivers, which can lead to inaccurate predictions and results. Therefore, ensuring that each data point is accurately labeled with its corresponding driver ID is critical to building a reliable and effective driver behavior model.

- **Preprocess data:**

In order to prepare the collected data for LLM input, it is important to preprocess it to ensure that it is in a suitable format. This involves several steps, including tokenization and conversion of data into sequences or batches. Tokenization refers to the process of breaking down the data into smaller units, such as words or phrases, which can then be analyzed by the LLM. This is particularly important for textual data, but can also be applied to numerical data, which may need to be segmented into appropriate units. Once the data has been tokenized, it can be converted into sequences or batches, which are essentially groups of data points that can be processed by the LLM in parallel. This can help to improve the speed and efficiency of the model, as well as ensure that the input data is properly formatted and aligned with the LLM's requirements. Overall, preprocessing the collected data is an essential step in

preparing it for LLM input, and can help to ensure that the model is able to analyze the data accurately and effectively.

- Numerical data: When working with numerical data, it is crucial to ensure that the data is clean and free of any outliers or anomalies that may lead to inaccurate results. Therefore, it is recommended to remove any such data points before proceeding with analysis. Additionally, to ensure consistency, it is helpful to normalize the values within the data set. This step can be followed by extracting meaningful features or creating time-series representations for better model interpretation. By doing so, one can obtain a better understanding of the data and make more informed decisions.

- Voice data: One of the most common tasks in modern speech technology is converting voice recordings into text. This process is carried out through Automatic Speech Recognition (ASR) technology, which uses complex algorithms to transcribe spoken language into written words. Additionally, it is possible to extract specific audio features from the recordings that may be relevant to the task at hand, such as tone or pitch. This information can be used to further analyze the speech data and gain insights into the speaker's emotions or intentions..

- Text data: To prepare text data for NLP, it's essential to apply several pre-processing techniques. These techniques include removing any irrelevant or noisy information from the text, correcting any spelling or grammatical errors, and tokenizing the text to split it into individual words. Additionally, normalization techniques such as stemming or lemmatization can be applied to transform words to their base form and reduce the number of unique words in the text. By performing these pre-processing steps, the text data is more suitable for further analysis using NLP techniques..

- Image data: When working with image data, it is often necessary to ensure uniformity in terms of size and aspect ratio. This can be achieved through techniques such as image resizing and cropping. In addition, it is important to normalize pixel values to ensure that they fall within a specific range. This can be done by subtracting the mean pixel value from each pixel and dividing by the standard deviation. To extract relevant features from images, techniques such as facial recognition or object detection can be employed. Facial recognition can be used to identify specific

individuals within an image or to detect facial expressions. Object detection can be used to identify specific objects within an image, such as cars or trees. These techniques can be used to extract features that can be used in further analysis or to train ML models.

8.1.2 Data Encoding

- Represent all modalities as text:

- Numerical data: When dealing with data analysis, one can use a technique called feature vector encoding. This method involves transforming data points into a set of numerical values that can be easily processed by algorithms. Alternatively, one can generate a statistical description of the data that can be interpreted as text. This method involves summarizing the data using statistical measures such as mean, median, and standard deviation, and then presenting the results in a textual form. Both of these techniques can be useful in different scenarios, depending on the nature of the data and the specific goals of the analysis.

- Voice data: Automated Speech Recognition (ASR) transcripts are computer-generated transcriptions of spoken language. They are widely used in various scenarios, such as generating subtitles for videos, transcribing interviews or meetings, and enabling voice-to-text features in smartphones and other devices.

ASR technology works by analyzing the audio input and converting it into written text. However, the accuracy of ASR transcripts can vary depending on several factors, such as the clarity of speech, background noise, the quality of the ASR technology, and the complexity of the language used. For example, ASR technology may struggle to accurately recognize accents or dialects that are different from the standard language model.

Similarly, the presence of background noise or overlapping speech can cause errors in the transcription. The quality of ASR technology can also vary depending on the vendor and the type of algorithm used. Some algorithms may be better suited for certain languages or accents than others. Therefore, it's important to carefully evaluate the accuracy of ASR transcripts before using them in critical applications.

- Text data: The text data provided is already preprocessed, which means that it has undergone a series of cleaning and tokenization processes. Cleaning refers to the removal of unwanted characters, such as special symbols, punctuation marks, and stop words,

while tokenization involves breaking down the text into smaller units, such as words or phrases, to facilitate analysis or ML tasks. Hence, the given text is ready for further processing or analysis without any additional cleaning or preprocessing steps.

– Image data: Image captioning is a computer vision task that involves generating a textual description of an image. It utilizes ML models to analyze visual features of an image and then generates a sentence or a paragraph that describes the contents of the image in a natural language format. By using outputs from an image captioning model, it is possible to transform visual data into textual descriptions. This can be especially useful in applications such as image search, robotics, and assistive technologies for individuals with visual impairments.

8.1.3 LLM as Driver Identification Model

- Choose a suitable LLM: When deciding on an LLM model, it's important to choose one capable of processing diverse textual inputs and encoding complex patterns. This is particularly important when dealing with large amounts of data or complex language structures. To ensure that you're selecting the most appropriate model for your needs, consider looking for models that are either inherently multimodal or have strong text classification capabilities. Multimodal models can process multiple input forms, such as text, images, and audio, while text classification models are specifically designed to categorize and analyze text data accurately. By evaluating these options carefully, you can feel confident that you're selecting an LLM model that will best meet your needs and produce the most accurate and valuable results.

- Fine-tune the LLM: To achieve optimal performance of the LLM, we can fine-tune it to adapt to our specific use case. This can be done by training the LLM on the encoded multimodal data while adjusting various hyperparameters, training strategies, and data sampling techniques. By doing so, we can customize the LLM to better suit our specific needs and improve its accuracy and efficiency. Keep in mind that the process of fine-tuning may require some experimentation and iteration to find the best combination of settings for our particular use case.

8.1.4 Evaluation and Refinement

Measure performance: When it comes to measuring the performance of a model, it's important to use metrics that provide a comprehensive evaluation. These metrics

should include accuracy, precision, recall, and F1-score. By evaluating the performance of the model across different data modalities and among various driver groups, we can ensure that the model is both robust and accurate. This allows us to identify any areas where the model may be struggling and make improvements to increase its overall effectiveness. Additionally, by measuring performance across different data modalities and driver groups, we can ensure that the model is capable of handling a wide range of scenarios and that it is not biased towards any particular group or situation. Overall, using a variety of metrics to evaluate model performance is crucial for ensuring that the model is effective and reliable.

9 LLM Application's for Driver Identification

LLM like GPT-4, developed by OpenAI, is an advanced version of the Generative Pre-trained Transformer (GPT) series^[129], exhibit advancements in NLP capabilities. With a potentially larger number of parameters, GPT-4 further enhances its contextual understanding, generation abilities, and application to various tasks. It could be utilized for driver identification in various applications. Leveraging its advanced language understanding and multimodal analysis capabilities, LLM can analyze driver identification data, including text, voice, and image inputs, to verify and authenticate the identity of drivers. By processing diverse data sources, LLM could contribute to enhance accuracy and reliability in driver identification processes, aiding in access control, personalized settings, driving behavior analysis, and security measures. This section explains the use of LLM's in driver identification.

9.1 Law Enforcement

9.1.1 License Plate Recognition

LLM can analyze images or video feeds from surveillance cameras or police vehicles to recognize and interpret license plates. By leveraging its language understanding capabilities, the model could extract relevant information from the license plates, such as vehicle registration details or owner information, aiding in driver identification.

9.1.2 Voice Analysis

LLM has the capability to analyze and identify distinct voice patterns or characteristics present in audio recordings. This feature can be utilized for the purpose

of matching a driver's voice with those of already known individuals or to provide supplementary evidence for identification purposes.

9.1.3 Integration with Law Enforcement Databases

The integration of LLM with law enforcement databases, such as criminal records or watchlists, enables the model to access and analyze this information. This analysis can help the model identify drivers who have previous records or are associated with ongoing investigations.

9.2 Usage-Based Insurance

9.2.1 Personalized Insurance Plans

Insurance companies could benefit from LLM's ability to create custom insurance plans for drivers based on their driving behavior, historical data, and other relevant factors. The model has the potential to assist in matching coverage and premiums to each driver's specific needs and risks.

9.2.2 Risk Scoring and Premium Calculation

LLM has the ability to generate personalized risk scores for drivers based on their driving behavior analysis. These scores can be utilized to establish insurance premiums that are more precise, reflecting the driver's real conduct on the road.

9.2.3 Claims Assessment

LLM is capable of providing assistance in evaluating the circumstances and variables which have resulted in an insurance claim. With the help of analyzing telematics data and other pertinent information, the system can assist in determining the responsibility, evaluating the extent of the incident, and simplifying the process of filing a claim.

9.2.4 Driving Insights and Feedback

Drivers can receive feedback and insights in real-time through LLM based on their behavior. Based on positive behavior changes, LLM can make personalized recommendations for improving driving skills, reducing risk, and even lowering insurance premiums.

9.2.5 Fraud Detection

The detection of fraudulent activities in UBI programs is something that LLM can assist with. Through analysis of the telematics data, the model is able to identify patterns, discrepancies, and anomalies that may point to instances of fraudulent behavior. For example, the tampering with or falsification of driving data could be detected using this approach.

9.3 Ride Sharing/E-hailing Industry

9.3.1 Driver Verification

LLM has the capability to provide more advanced verification mechanisms which would only allow verified and authorized drivers to participate in the ride-sharing platform. The platform can examine and authenticate the credentials, licenses, and background checks of drivers, thereby improving the verification process.

9.3.2 Biometric Authentication

It is possible to integrate biometric authentication techniques, such as voice recognition, into LLM for driver identification purposes. This would involve comparing the driver's biometric information with previously registered profiles, thus adding an extra layer of security to prevent unauthorized access to the platform.

9.3.3 Anti-Fraud Measures

The ride-sharing platform can leverage LLM to detect and prevent fraudulent activities. LLM can analyze driver behavior or transaction data to identify patterns, discrepancies, and anomalies, which can signal potential cases of fraudulent behavior like identity theft or account manipulation.

9.3.4 Ride Verification

LLM has the ability to help verify that rides are completed and the correct driver is assigned to each trip. The model uses trip data, geolocation information, and driver identity to cross-check the information for preventing unauthorized ride pickups or fraudulent claims.

9.3.5 Driver Ratings and Feedback

Analyzing feedback from passengers can be a valuable way for LLM to contribute to the driver rating system. By analyzing text data, the model can extract sentiments, identify patterns, and provide insights that can help drivers improve their service quality.

9.4 ADAS

Drivers can receive real-time feedback on driving information, vehicle status, and other pertinent details through text and voice communication from their vehicles. This increased interaction between drivers and their vehicles enhances the overall driving experience. LLM can also contribute to this by analyzing driver behavior data, allowing drivers to gain insights into their driving behavior and receive personalized driving advice. As a result, driving safety

is improved, and a more informed and responsible approach to driving is promoted.

9.4.1 Driver Monitoring

The utilization of LLM is possible for monitoring and identifying the driver's state and attentiveness while driving. Through the analysis of voice patterns, this model can determine whether the driver is engaged, fatigued, or distracted, thus contributing to safer driving.

9.4.2 Personalized Settings

ADAS settings can be personalized by LLM based on driver identification. The system can automatically adjust seat position, mirrors, climate control, and other preferences to match the identified driver's profile once the driver is recognized.

9.4.3 Driver-Specific Training and Guidance

LLM has the capability to offer customized training and instruction to drivers by analyzing their identification and driving habits. By examining driving data, the system can detect areas that require improvement and suggest real-time feedback or recommendations to improve driving abilities and ensure safety.

9.5 Fleet Management

9.5.1 Driver Authentication

LLM has the capability to provide advanced authentication mechanisms that guarantee that only authorized drivers can gain access to and operate fleet vehicles. The model can examine driver identification data, including biometric information or credentials, to authenticate the identity of drivers prior to permitting them access to vehicles.

9.5.2 Driving Behavior Analysis

Fleet vehicles' driving behavior data can be analyzed by LLM to assess driver performance and safety. By analyzing telematics data that includes speed, acceleration, braking, and adherence to traffic rules, the model can evaluate individual driver behavior and give suggestions for improving performance or reducing risks.

9.5.3 Safety Monitoring

Real-time monitoring of driver safety within fleet vehicles can be enhanced by LLM. The model has the ability to analyze sensor data, driver behaviour, and environmental factors to detect potential risks or hazards. In case of any issues, the model promptly sends alerts or interventions to prevent accidents and enhance overall safety.

9.5.4 Compliance Monitoring

Fleet management can benefit from LLM's assistance in ensuring compliance with regulations and company policies. LLM's analysis of driver behavior and operational data can detect instances of non-compliance, such as extended idling or speeding, and generate notifications or reports to ensure adherence to regulations and guidelines.

9.5.5 Incident Investigation

LLM can help investigate and analyze any accidents or incidents involving fleet vehicles. The model can analyze telematics data, driver behavior, and external factors to provide insights into the cause and contributing factors of the accident. These insights can be helpful in processing insurance claims and taking preventive measures to avoid future incidents.

9.6 Automotive Security

9.6.1 User Authentication

LLM has the capability to aid in the implementation of strong user authentication systems in automobiles. To make certain that only authorized individuals are able to access and operate the vehicle's systems, it can utilize various biometric authentication methods such as voice recognition, facial recognition, and others.

9.6.2 Anomaly Detection

During the process of driver identification, LLM has the capability to recognize and alert about possible anomalies or suspicious activity. In order to do so, the model analyzes the present driver's traits and actions, comparing them with previous data to identify any unusual patterns that may suggest security threats or unauthorized entry.

9.6.3 Integration with Security Systems

It is possible to combine LLM with various other automotive security systems, including intrusion detection systems, sensors, and surveillance cameras. The model's language comprehension and analysis capabilities can be utilized to link driver identification data with other security events, resulting in a comprehensive security solution.

9.7 Strengths of LLM

9.7.1 Language Understanding

LLM has the ability to understand language at an advanced level, which enables it to analyze and comprehend driver identification information in

different formats, including text, voice, and images. This feature can facilitate more precise and all-encompassing driver identification.

9.7.2 Multimodal Analysis

LLM has the ability to combine various data sources such as biometric information, driving behavior data, and identification documents. This integration enables the model to improve the precision and dependability of driver identification by comprehensively analyzing the different types of data available.

9.7.3 Continuous Learning

The LLM model can consistently learn and adjust to novel driver identification patterns and approaches. This feature may facilitate the model to enhance its performance with time and remain up-to-date with evolving identification techniques or security measures.

9.8 Weaknesses of LLM

9.8.1 Dependency on Data Quality

The effectiveness of LLM, similar to other AI models, is dependent on the quality and variety of the data used in its training. In case the training data for driver identification is inadequate, prejudiced, or restricted, it may result in incorrect outcomes or partiality in the identification procedure.

9.8.2 Vulnerability to Adversarial Attacks

LLM is vulnerable to adversarial attacks because of intentional input manipulation by malicious individuals who aim to trick the model and bypass driver identification systems. To prevent such attacks, it's crucial to have strong security measures and adequate safeguards in place.

9.8.3 Ethical and Privacy Concerns

There are ethical and privacy concerns associated with using LLM or any AI model for driver identification. To ensure responsible and legal use of driver identification technologies, it is important to implement proper measures for consent, transparency, and data protection.

9.8.4 Hardware and Processing Requirements

Deploying LLM, which is a more advanced model, can be challenging in resource-constrained environments or devices that have limited capabilities due to the significant computational resources and processing power it requires.

10 Performance evaluation

It is essential to choose appropriate performance

measures for classification systems based on their specific domain^[130]. One common approach for evaluating classification performance is to use a confusion matrix, which assesses a classification problem using test data. The confusion matrix is useful in determining the prediction accuracy for both negative and positive class instances. A True Positive (TP) is when both the actual and predicted classes are positive, indicating accurate prediction of positives. A True Negative (TN) is when both the predicted and actual classes are negative. False Positive (FP) cases are when the actual class is negative, but the expected class is positive, whereas False Negative (FN) cases are when the actual category is positive, but the projected category is negative. To optimize the performance of a driver identification system, it is crucial to minimize both FN and FP.

In driver identification systems, FP and FN serve as critical performance indicators, especially in contexts that demand high levels of security and reliability.

- **False Positives (FP):** This occurs when an unauthorized driver is wrongly identified as an authorized one. A high rate of FP can expose significant security risks, potentially granting unauthorized access to vehicles, which is particularly problematic in areas like fleet management, autonomous vehicles, and usage-based insurance.

- **False Negatives (FN):** This refers to the situation where an authorized driver is incorrectly denied access. Frequent instances of FN can harm user experience, leading to frustration for legitimate users who may face repeated authentication challenges, which could discourage them from embracing the system.

Design Strategies to Optimize FP and FN

- **Feature Selection and Engineering:** Selecting specific driving behavior characteristics (such as acceleration patterns, steering techniques, and braking habits) can significantly boost classification accuracy and help reduce misclassification rates.

- **Threshold Optimization:** Fine-tuning decision thresholds within classification models can help strike a balance between FP and FN rates. A lower threshold might decrease FN but raise FP, while a higher threshold tends to have the opposite effect. Thus, a careful trade-off is needed based on specific application needs.

- **Ensemble Learning and Hybrid Models:** Merging different ML approaches (like combining DL with rule-based models) can enhance system

Table 8 Frequency of performance evaluation used in review studies. $\checkmark$ and $\times$ indicates inclusion and exclusion respectively.

References	Accuracy	Precision	Recall	F1 Score	ERR	ROC
[10, 27, 30, 36, 39, 69, 71, 83, 89, 90, 96]	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\times$
[4, 18, 31, 38, 41, 70, 87]	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\times$
[16, 21, 23, 81]	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
[28]	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
[52]	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
[13]	$\times$	$\times$	$\times$	$\checkmark$	$\times$	$\times$
[49]	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
[43, 85]	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
[82, 88]	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$
[79]		$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	
[86]	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	
[42, 51]	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	$\times$
[24]	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\times$
[32]	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	$\times$
[19, 20]	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	$\times$
[69]	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$
[29]	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\times$
[86]	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	$\times$

robustness and lower error rates by using their varying strengths.

- **Personalized Models:** Creating models tailored to individual drivers through transfer learning or adaptive algorithms can fine-tune the system to better match personal driving styles, resulting in fewer FN cases for genuine drivers.

- **Continuous Authentication Mechanisms:** Rather than depending solely on one-time authentication, constantly monitoring driving behavior over time can help alleviate both FP and FN issues, leading to a more dynamic and responsive identification system.

Accuracy, recall, precision, and F-measure are some of the performance metrics frequently used in driver identification to assess classifier performance. Table 8 shows all the evaluation metrics used by the researchers. Descriptions of these performance indicators are provided below.

10.1 Accuracy

Driver identification is often evaluated using accuracy as a metric to determine correctness. This metric measures the proportion of correctly identified drivers

to the total sample size. However, accuracy alone might not accurately identify imbalances in the dataset, especially if the classes are not evenly distributed.

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (12)$$

10.2 Precision

Precision evaluates the model's capability to accurately identify drivers among all the positive identifications made. It focuses on minimizing false positives.

$$Precision = (TP) / ((TP + FP)) \quad (13)$$

10.3 Recall

Recall is a measure used to evaluate the accuracy of a model in identifying positive cases (drivers) correctly. This measure aims to minimize the number of false negatives.

$$Recall = TP / ((TP + FN)) \quad (14)$$

10.4 F1 Score

The F1 score is a metric that considers both precision and recall, providing a comprehensive evaluation of a model's performance in terms of false negatives and false positives.

$$F1Score = 2 \times (Precision * Recall) / (Precision + Recall) \quad (15)$$

10.5 Equal Error Rate (EER)

Biometric systems, such as driver identification, commonly use EER as a performance evaluation metric. EER represents the point of balance between false acceptance (FAR) and false rejection (FRR) rates in the system's performance. For driver identification specifically, EER indicates the threshold where the system's identification errors for different drivers are in equilibrium. To calculate FAR, FRR, and EER, we can use the following formulas:

$$FAR = FPR = FP / ((FP + TN)) \quad (16)$$

$$FRR = FNR = FN / ((FN + TP)) \quad (17)$$

$$EER = ((FAR + FRR)) / 2 \quad (18)$$

10.6 ROC Curve and AUC

The graphical representation of the performance of a model across different discrimination thresholds is provided by the Receiver Operating Characteristic (ROC) curve. Meanwhile, the scalar value that quantifies this performance is known as the Area

Under the Curve (AUC). The ROC curve and AUC provide a quantification of the balance between the sensitivity (true positive rate) and specificity (false positive rate) of the model.

11 Future research directions

Several research challenges were identified in the driver identification sector through a review study. To enhance the effectiveness of driver identification systems, substantial research efforts are required to address the highlighted research topic. Below, we delve deeper into the open research issues that need to be tackled.

(1) **Robustness and Generalization:** Improving the resilience of driver identification models to cope with varying driving conditions, road types, and environmental elements. Advancing the overall adaptability of models to precisely recognize drivers in different situations and datasets.

(2) **Real-Time and Online Systems:** Real-time and online driver identification systems are being developed to process data instantly and detect drivers and possible anomalies immediately. This necessitates the optimization of algorithms and architectures to achieve high processing speed with minimal latency.

(3) **Multimodal Data Fusion:** Utilizing the fusion of multiple data modalities, including visual data, audio data, and physiological signals such as heart rate and eye movement, can improve the precision and dependability of driver identification. By merging data from different sensors and sources, it is possible to obtain a more thorough comprehension of driver behavior.

(4) **Privacy and Security:** Ensuring the security of driver identification systems and addressing privacy concerns. Creating strategies to safeguard sensitive personal data and prohibit unauthorized access or misuse of driver information.

(5) **Unsupervised and Semi-supervised Learning:** Exploring unsupervised and semi-supervised learning methods for identifying drivers, even when there is a lack of labeled data due to cost or limited availability. These techniques can help reduce the reliance on large, labeled datasets and make driver identification more feasible in practical situations.

(6) **Transfer Learning and Domain Adaptation:** Investigating transfer learning and domain adaptation techniques to allow the transfer of knowledge from one driving situation to another. This could aid in the

creation of driver identification models that are effective in a variety of driving scenarios, regardless of the type of vehicle, demographic, or environment.

(7) Ethical and Legal Considerations:

Developing ethical and legal strategies to handle driver identification issues, including data privacy, consent, and avoiding any potential biases. Establishing frameworks and guidelines to ensure using driver identification technologies in a fair and responsible manner.

(8) **Real-World Deployment and Validation:** The process of carrying out comprehensive field trials and validation studies to evaluate the effectiveness and practicality of driver identification systems. Our aim is to assess these systems' real-world applicability and determine their scalability, reliability, and usability in practical settings. This evaluation will help us integrate these systems into current transportation systems and applications.

(9) **LLM-Based Approaches for Driver Identification:** Future research should explore the unique benefits of LLMs over traditional DL methods. Specifically, investigations should focus on:

- **Comparative Advantages:** Evaluating LLMs' enhanced contextual understanding and few-shot learning capabilities compared to conventional models.
- **Domain-Specific Design:** Developing strategies for designing and fine-tuning domain-specific LLMs tailored for driver identification.
- **Practical Challenges:** Addressing computational resource requirements, privacy concerns, and ensuring robust model generalization in diverse driving conditions.
- **Experimental Validation:** Implementing and validating these approaches through comprehensive experiments and providing corresponding code implementations in future studies.

12 Lessons Learned

Throughout the course of this comprehensive review, several critical lessons have emerged that can significantly influence the future direction of driver identification research and its practical applications.

• Importance of Data Quality and Diversity:

One of the primary lessons learned is the crucial role of data quality and diversity in developing robust driver identification systems. High-quality, diverse datasets that encompass a wide range of driving conditions, behaviors, and environmental factors are

essential for training models that generalize well across different scenarios. Future research should prioritize the collection and utilization of such datasets to enhance the performance and reliability of driver identification systems.

- **Integration of Multimodal Data:**

The integration of multimodal data, including visual, audio, and physiological signals, has shown to improve the precision and dependability of driver identification. This approach allows for a more holistic understanding of driver behavior and can significantly enhance system accuracy. Researchers should continue to explore and refine techniques for effectively fusing data from various sources.

- **Real-Time Processing and System Efficiency:**

Real-time processing capabilities are vital for the practical deployment of driver identification systems. The development of algorithms and architectures that can handle data processing with minimal latency is crucial for real-world applications. Lessons from current studies highlight the need for optimizing computational efficiency without compromising accuracy.

- **Adaptability and Generalization:**

Driver identification models must be adaptable and generalizable to various driving contexts and conditions. The ability to transfer knowledge and adapt to new environments is essential for the widespread adoption of these systems. Research should focus on developing models with strong generalization capabilities, using techniques like transfer learning and domain adaptation.

By learning from these insights, researchers and practitioners can develop more effective, reliable, and ethically sound driver identification systems. These lessons will guide future innovations and contribute to the advancement of automotive safety and personalization technologies.

13 Conclusion

In this paper, we have conducted a comprehensive examination of the components integral to driver identification systems. Our exploration covered various data sources, including CAN-BUS/OBD-II interfaces, smartphones, GPS, IMUs, and wearable technologies, analyzing their strengths and limitations. Preprocessing data from these sources is critical for developing robust identification systems. We examined multiple datasets based on data source, average trip, road type, and data

collection time. Feature engineering methodologies were strategically applied to optimize accuracy, speed up computational processes, and minimize classification errors. We selected a classification algorithm from a range of ML and DL options to ensure accurate results. Our analysis highlighted the efficacy of hybrid models in intelligently selecting algorithms tailored to specific objectives and data availability, enhancing generalization and resilience. We also explored the potential of advanced LLM frameworks in driver identification, highlighting their revolutionary potential. Evaluation matrices used in existing studies were identified for maintaining generalization.

The broader significance of this research lies in its potential to enhance automotive security, law enforcement, ride-sharing, ADAS, fleet management, and usage-based insurance. Improved driver identification systems can lead to safer and more customized driving experiences, reducing accidents caused by driver error.

Future research should focus on enhancing model robustness and generalization, developing real-time systems for instant data processing and anomaly detection, and integrating multimodal data to improve precision and reliability. Ensuring privacy and security is crucial, requiring strategies to protect sensitive data. Exploring unsupervised and semi-supervised learning can reduce reliance on large labeled datasets, making identification feasible with limited data. Transfer learning and domain adaptation can enable models to be effective across diverse scenarios. Addressing ethical and legal considerations is vital, requiring frameworks for data privacy, consent, and biases. Real-world validation studies will assess the practicality and scalability of these systems, ensuring their effective integration into current transportation applications.

By pursuing these research directions, the field of driver identification can continue to evolve, addressing current limitations and leveraging new technological advancements. This evolution will contribute to developing more secure and efficient transportation systems, enhancing driver safety and personalization on a broader scale.

Acknowledgment

The authors would like to acknowledge the support of Prince Sultan University for paying the article processing charges (APC) of this manuscript.

References

- [1] Z. Mahmood, T. Ali, S. Khattak, S. U. Khan, and L. T. Yang, "Automatic vehicle detection and driver identification framework for secure vehicle parking," in *2015 13th International Conference on Frontiers of Information Technology (FIT)*. IEEE, 2015, pp. 6–11.
- [2] A. Burton, T. Parikh, S. Mascarenhas, J. Zhang, J. Voris, N. S. Artan, and W. Li, "Driver identification and authentication with active behavior modeling," in *2016 12th International Conference on Network and Service Management (CNSM)*. IEEE, 2016, pp. 388–393.
- [3] R. Chen, M. F. She, X. Sun, L. Kong, and Y. Wu, "Driver recognition based on dynamic handgrip pattern on steering wheel," in *2011 12th ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing*. IEEE, 2011, pp. 107–112.
- [4] P.-Y. Tseng, P.-C. Lin, and E. Kristianto, "Vehicle theft detection by generative adversarial networks on driving behavior," *Engineering Applications of Artificial Intelligence*, vol. 117, 2023.[Online]. Available: [GotoISI : //WOS: 000894963300007](https://www.sciencedirect.com/science/article/pii/S0950068723000007)
- [5] E. Romera, L. M. Bergasa, and R. Arroyo, "A realtime multi-scale vehicle detection and tracking approach for smartphones," in *ITSC, Conference Proceedings*, pp. 1298–1303.
- [6] F. B. Labs, "Openxc database," 2020.[Online]. Available: <http://openxcplatform.com/resources/traces.html>
- [7] Z. Khan, P. Fan, S. Fang, and F. Abbas, "An unsupervised cluster-based vanet-oriented evolving graph (cvoeg) model and associated reliable routing scheme," *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, no. 10, pp. 3844–3859, 2019.
- [8] S. Sarala, D. S. Yadav, and A. Ansari, "Emotionally adaptive driver voice alert system for advanced driver assistance system (adas) applications," in *2018 International Conference on Smart Systems and Inventive Technology (ICSSIT)*. IEEE, 2018, pp. 509–512.
- [9] C. M. Martinez, M. Heucke, F.-Y. Wang, B. Gao, and D. Cao, "Driving style recognition for intelligent vehicle control and advanced driver assistance: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 19, no. 3, pp. 666–676, 2017.
- [10] A. H. Mekki, A. Bouhout, and I. Berrada, "Improving driver identification for the next-generation of in-vehicle software systems," *Ieee Transactions on Vehicular Technology*, vol. 68, no. 8, pp. 7406–7415, 2019.[Online]. Available: [GotoISI : //WOS: 000481944300014](https://ieeexplore.ieee.org/document/8746156) [https://ieeexplore.ieee.org/document/8746156/](https://ieeexplore.ieee.org/document/8746156)
- [11] S. Taamneh, P. Tsiamyrtzis, M. Dcosta, P. Buddharaju, A. Khatri, M. Manser, T. Ferris, R. Wunderlich, and I. Pavlidis, "A multimodal dataset for various forms of distracted driving," *Scientific data*, vol. 4, no. 1, pp. 1–21, 2017.
- [12] E. Romera, L. M. Bergasa, and R. Arroyo, "Need data for driver behaviour analysis? presenting the public uahdriveset," in *2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, Conference Proceedings, pp. 387–392.
- [13] H. Hu, J. Liu, Z. Gao, and P. Wang, "Driver identification using 1d convolutional neural networks with vehicular can signals," *Iet Intelligent Transport Systems*, vol. 14, no. 13, pp. 1799–1809, 2020.[Online]. Available: [GotoISI : //WOS: 000619167800007](https://www.sciencedirect.com/science/article/pii/S1751975320300007)
- [14] Y. Xun, J. Liu, and Y. Zhang, "Side-channel analysis for intelligent and connected vehicle security: A new perspective," *Ieee Network*, vol. 34, no. 2, pp. 150–157, 2020.[Online]. Available: [GotoISI : //WOS: 000555430800022](https://www.sciencedirect.com/science/article/pii/S1045911320300022) <https://ieeexplore.ieee.org/document/8839975/>
- [15] B. I. Kwak, M. L. Han, and H. K. Kim, "Driver identification based on wavelet transform using driving patterns," *Ieee Transactions on Industrial Informatics*, vol. 17, no. 4, pp. 2400–2410, 2021.[Online]. Available: [GotoISI : //WOS:000607814600011](https://www.sciencedirect.com/science/article/pii/S1568484621300011) <https://ieeexplore.ieee.org/document/9108547/>
- [16] S. Hernandez Sanchez, R. Fernandez Pozo, and L. A. Hernandez Gomez, "Driver identification and verification from smartphone accelerometers using deep neural networks," *Ieee Transactions on Intelligent Transportation Systems*, vol. 23, no. 1, pp. 97–109, 2022.[Online]. Available: [GotoISI : //WOS: 000735517000010](https://www.sciencedirect.com/science/article/pii/S1568484622300010) <https://ieeexplore.ieee.org/document/9145829/>
- [17] B. I. Kwak, J. Woo, and H. K. Kim, "Know your master: Driver profiling-based anti-theft method," in *2016 14th Annual Conference on Privacy, Security and Trust (PST)*. IEEE, Conference Proceedings, pp. 211–218.
- [18] N. Virojboonkiate, A. Chanakitkarnchok, P. Vateekul, and K. Rojviboonchai, "Public transport driver identification system using histogram of acceleration data," *Journal of Advanced Transportation*, 2019.[Online]. Available: [GotoISI : //WOS:000459036200001](https://www.hindawi.com/journals/jat/2019/6372597.pdf) [https://downloads.hindawi.com/journals/jat/2019/6372597.pdf](https://www.hindawi.com/journals/jat/2019/6372597.pdf)
- [19] K. Dolos, C. Meyer, A. Attenberger, and J. Steinberger, "Driver identification using in-vehicle digital data in the forensic context of a hit and run accident," *Forensic Science International–Digital Investigation*, vol. 35, 2020, times Cited: 5052666–2817.[Online]. Available: [GotoISI : //WOS: 000600551900018](https://www.sciencedirect.com/science/article/pii/S1568484620300018)
- [20] —, "Forensic driver identification considering an unknown suspect," *International Journal of Applied Mathematics and Computer Science*, vol. 31, no. 4, pp. 587–599, 2021.[Online]. Available: [GotoISI : //WOS: 000740632100004](https://www.sciencedirect.com/science/article/pii/S104591132100004) <https://sciencedirect.com/science/article/pii/S104591132100004>
- [21] O. Abul and B. Karatas, "Can driving patterns predict identity and gender?" *Journal of Ambient Intelligence and Humanized Computing*, vol. 12, no. 1, pp. 151–166, 2021.[Online]. Available: [GotoISI : //WOS:](https://www.sciencedirect.com/science/article/pii/S1568484621300011)

- 000619874500013<https://link.springer.com/article/10.1007/s12652-019-01457-1>
- [22] K. Takeda, J. H. Hansen, P. Boyraz, L. Malta, C. Miyajima, and H. Abut, International large-scale vehicle corpora for research on driver behavior on the road, *IEEE Transactions on Intelligent Transportation Systems*, vol. 12, no. 4, pp. 1609–1623, 2011.
- [23] Y. Xun, J. Liu, and Z. Shi, “Multitask learning assisted driver identity authentication and driving behavior evaluation,” *Ieee Transactions on Industrial Informatics*, vol. 17, no. 10, pp. 7093–7102, 2021.[Online]. Available: [⟨GotoISI⟩ : //WOS:000673414500052https://ieeexplore.ieee.org/document/9244617/](https://ieeexplore.ieee.org/document/9244617)
- [24] Y. Ma, J. Wu, C. Long, and Y.-B. Lin, “Mobidiv: A privacy-aware real-time driver identity verification on mobile phone,” *Ieee Internet of Things Journal*, vol. 9, no. 4, pp. 2802–2816, 2022.[Online]. Available: [⟨GotoISI⟩ : //WOS:000752017900036https://ieeexplore.ieee.org/document/9467376/](https://ieeexplore.ieee.org/document/9467376/)
- [25] A. El Mekki, A. Bouhoute, and I. Berrada, Improving driver identification for the next-generation of in-vehicle software systems, *IEEE Transactions on Vehicular Technology*, vol. 68, no. 8, pp. 7406–7415, 2019.
- [26] D. M. Cades, C. Crump, B. D. Lester, and D. Young, “Driver distraction and advanced vehicle assistive systems (adas): investigating effects on driver behavior,” in *Advances in Human Aspects of Transportation: Proceedings of the AHFE 2016 International Conference on Human Factors in Transportation, July 27–31, 2016, Walt Disney World®, Florida, USA*. Springer, 2017, pp. 1015–1022.
- [27] G. H. Choi, K. Lim, and S. B. Pan, “Driver identification system using normalized electrocardiogram based on adaptive threshold filter for intelligent vehicles,” *Sensors*, vol. 21, no. 1, 2021.[Online]. Available: [⟨GotoISI⟩ : //WOS: 000606145800001https://mdpi-res.com/d_attachment/sensors/sensors-21-00202/article_deploy/sensors-21-00202-v2.pdf?version=1609826259](https://mdpi-res.com/d_attachment/sensors/sensors-21-00202/article_deploy/sensors-21-00202-v2.pdf?version=1609826259)
- [28] G. Balan, S. Arumugam, S. Muthusamy, H. Panchal, H. Kotb, M. Bajaj, S. S. M. Ghoneim, and Kitmo, “An improved deep learning-based technique for driver detection and driver assistance in electric vehicles with better performance,” *International Transactions on Electrical Energy Systems*, vol. 2022, 2022.[Online]. Available: [⟨GotoISI⟩ : //WOS: 000884986400001https://downloads.hindawi.com/journals/itees/2022/8548172.pdf](https://downloads.hindawi.com/journals/itees/2022/8548172.pdf)
- [29] M. L. Bernardi, M. Cimitile, F. Martinelli, and F. Mercaldo, “Driver and path detection through time-series classification,” *Journal of Advanced Transportation*, 2018.[Online]. Available: [⟨GotoISI⟩ : //WOS: 000428908200001https://downloads.hindawi.com/journals/jat/2018/1758731.pdf](https://downloads.hindawi.com/journals/jat/2018/1758731.pdf)
- [30] L. Moreira-Matias and H. Farah, “On developing a driver identification methodology using invehicle data recorders,” *Ieee Transactions on Intelligent Transportation Systems*, vol. 18, no. 9, pp. 2387–2396, 2017.[Online]. Available: [⟨GotoISI⟩ : //WOS: 000408755500011https://ieeexplore.ieee.org/document/7819486/](https://ieeexplore.ieee.org/document/7819486/)
- [31] T. Oktay, E. Yogurtcuoglu, R. N. Sarikaya, A. R. Karaca, M. F. Komurcu, and A. Sayar, “Multimodel anomaly detection on spatio-temporal logistic datastream with open anomaly detection architecture,” *Expert Systems with Applications*, vol. 186, 2021.[Online]. Available: [⟨GotoISI⟩ : //WOS: 000703905700010](https://ieeexplore.ieee.org/document/7819486/)
- [32] M. N. Azadani and A. Boukerche, “Driverrep: Driver identification through driving behavior embeddings,” *Journal of Parallel and Distributed Computing*, vol. 162, pp. 105–117, 2022.[Online]. Available: [⟨GotoISI⟩ : //WOS: 000777805100006](https://ieeexplore.ieee.org/document/7819486/)
- [33] D. Yi, Z. Lei, S. Liao, and S. Z. Li, “Learning face representation from scratch,” *arXiv preprint arXiv:1411.7923*, 2014.
- [34] S. Schneegass, B. Pfleging, N. Broy, F. Heinrich, and A. Schmidt, “A data set of real world driving to assess driver workload,” in *Proceedings of the 5th international conference on automotive user interfaces and interactive vehicular applications*, Conference Proceedings, pp. 150–157.
- [35] H. Abut, H. Erdoğan, A. Erçil, A. B. Çürüklü, H. C. Koman, F. Tas, A. Argunş ah, B. Akan, H. Karabalkan, and E. Çökelek, “Data collection with” uyanik”: too much pain; but gains are coming,” 2007.
- [36] G. H. Choi, K. Lim, and S. B. Pan, “Identification system based on resolution adjusted 2d spectrogram of driver’s ecg for intelligent vehicle,” *Mobile Information Systems*, vol. 2022, 2022.[Online]. Available: [⟨GotoISI⟩ : //WOS000769748600001https://downloads.hindawi.com/journals/misy/2022/5404343.pdf](https://downloads.hindawi.com/journals/misy/2022/5404343.pdf)
- [37] H. Hu, J. Liu, G. Chen, Y. Zhao, Y. Men, and P. Wang, “Driver identification through vehicular can bus data: An ensemble deep learning approach,” *Iet Intelligent Transport Systems*, 2022.[Online]. Available: [⟨GotoISI⟩ : //WOS: 000879124800001](https://ieeexplore.ieee.org/document/7819486/)
- [38] G. Priyadharshini and M. F. Ukrit, “An empirical evaluation of importance-based feature selection methods for the driver identification task using obd data,” *International Journal of System Assurance Engineering and Management*, 2022.[Online]. Available: [⟨GotoISI⟩ : //WOS: 000824916800001https://link.springer.com/article/10.1007/s13198-022-01695-1](https://link.springer.com/article/10.1007/s13198-022-01695-1)
- [39] Z. Ding, D. Xu, C. Tu, H. Zhao, M. Moze, F. Aioun, and F. Guillemard, “Driver identification through heterogeneity modeling in car-following sequences,” *Ieee Transactions on Intelligent Transportation Systems*, vol. 23, no. 10, pp. 17143–17156, 2022.[Online]. Available: [⟨GotoISI⟩ : //WOS:000764846100001https://ieeexplore.ieee.org/document/9724165/](https://ieeexplore.ieee.org/document/9724165/)
- [40] J. Li, Q. Zhou, Y. He, H. Williams, and H. Xu, “Driver-

- identified supervisory control system of hybrid electric vehicles based on spectrum-guided fuzzy feature extraction,” *Ieee Transactions on Fuzzy Systems*, vol. 28, no. 11, pp. 2691–2701, 2020.[Online]. Available: [〈GotoISI〉 ://WOS:000587703000003https://ieeexplore.ieee.org/document/8993840/](https://ieeexplore.ieee.org/document/8993840/)
- [41] R. Ahmadian, M. Ghatee, and J. Wahlstrom, “Discrete wavelet transform for generative adversarial network to identify drivers using gyroscope and accelerometer sensors,” *Ieee Sensors Journal*, vol. 22, no. 7, pp. 6879–6886, 2022.[Online]. Available: [〈GotoISI〉 ://WOS: 000792967000075https://ieeexplore.ieee.org/document/9716101/](https://ieeexplore.ieee.org/document/9716101/)
- [42] G. Ahmadi-Assalemi, H. M. Al-Khateeb, C. Maple, G. Epiphaniou, M. Hammoudeh, H. Jahankhani, and P. Pillai, “Optimising driver profiling through behaviour modelling of in-car sensor and global positioning system data,” *Computers and Electrical Engineering*, vol. 91, 2021.[Online]. Available: [〈GotoISI〉 ://WOS: 000691871700008](https://www.sciencedirect.com/science/article/pii/S0045791421000008)
- [43] C. Ravi, A. Tigga, G. T. Reddy, S. Hakak, and M. Alazab, “Driver identification using optimized deep learning model in smart transportation,” *ACM Transactions on Internet Technology*, vol. 22, no. 4, pp. 1–17, 2022.
- [44] Z. Khan, P. Fan, F. Abbas, H. Chen, and S. Fang, “Two-level cluster based routing scheme for 5g v2x communication,” *IEEE Access*, vol. 7, pp. 16194–16205, 2019.
- [45] Z. Khan and P. Fan, “A multi-hop moving zone (mmz) clustering scheme based on cellular-v2x,” *China Communications*, vol. 15, no. 7, pp. 55–66, 2018.
- [46] Z. Khan, A. Koubaa, and H. Farman, “Smart route: Internet-of-vehicles (ioV)-based congestion detection and avoidance (ioV-based cda) using rerouting planning,” *Applied Sciences*, vol. 10, no. 13, p. 4541, 2020.
- [47] Z. Khan, S. Fang, A. Koubaa, P. Fan, F. Abbas, and H. Farman, “Street-centric routing scheme using ant colony optimization-based clustering for bus-based vehicular ad-hoc network,” *Computers and Electrical Engineering*, vol. 86, p. 106736, 2020.
- [48] W. Dong, J. Li, R. Yao, C. Li, T. Yuan, and L. Wang, “Characterizing driving styles with deep learning,” *arXiv preprint arXiv: 1607.03611*, 2016.
- [49] M. A. Rahim, J. Liu, Z. Zhang, L. Zhu, X. Li, and S. Khan, “Who is driving? event-driven driver identification and impostor detection through support vector machine,” *Ieee Sensors Journal*, vol. 20, no. 12, pp. 6552–6559, 2020.[Online]. Available: [〈GotoISI〉 ://WOS: 000536772100039https://ieeexplore.ieee.org/document/9018037/](https://ieeexplore.ieee.org/document/9018037/)
- [50] X. Yu, H. Zhao, L. Zhang, S. Wu, B. Krishnamachari, and V. O. Li, “Cooperative sensing and compression in vehicular sensor networks for urban monitoring,” in *2010 IEEE International Conference on Communications, IEEE, Conference Proceedings*, pp. 1–5.
- [51] R. R. Mardi Putri, C.-H. Yang, C.-C. Chang, and D. Liang, “Smartwatch-based open-set driver identification by using gmm-based behavior modeling approach,” *Ieee Sensors Journal*, vol. 21, no. 4, pp. 4918–4926, 2021.[Online]. Available: [〈GotoISI〉 ://WOS: 000611133100099https://ieeexplore.ieee.org/document/9222135/](https://ieeexplore.ieee.org/document/9222135/)
- [52] F. Martinelli, F. Mercaldo, A. Orlando, V. Nardone, A. Santone, and A. K. Sangaiah, “Human behavior characterization for driving style recognition in vehicle system,” *Computers and Electrical Engineering*, vol. 83, 2020.[Online]. Available: [〈GotoISI〉 ://WOS: 000530032500001](https://www.sciencedirect.com/science/article/pii/S0045791420300001)
- [53] M. Abu Tareq Rony, M. Shariful Islam, T. Sultan, S. Alshathri, and W. El-Shafai, “Exploring potentials of conventional and large language models on medical data,” *IEEE Access*, vol. 12, pp. 103473–103487, 2024.
- [54] M. Tounsi, A. J. Mahdi, M. A. Ahmed, A. T. Azar, D. A. Smaït, S. Ahmed, A. M. Zalzal, and I. K. Ibraheem, “Hardware implementation of a deep learning-based autonomous system for smart homes using field programmable gate array technology,” *Engineering, Technology and Applied Science Research*, Sep 2024.
- [55] S. A. El-Nabi, A. F. Ibrahim, E.-S. M. El-Rabaie, O. F. Hassan, N. F. Soliman, K. F. Ramadan, and W. El-Shafai, “Driver drowsiness detection using swin transformer and diffusion models for robust image denoising,” *IEEE Access*, vol. 13, pp. 71880–71907, 2025.
- [56] D. Zhao, J. Hou, Y. Zhong, W. He, Z. Fu, and F. Zhou, “Driver identification methods in electric vehicles, a review,” *World Electric Vehicle Journal*, vol. 13, no. 11, p. 207, 2022.
- [57] M. N. Azadani and A. Boukerche, “Driving behavior analysis guidelines for intelligent transportation systems,” *Ieee Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 6027–6045, 2022.[Online]. Available: [〈GotoISI〉 ://WOS: 000733552800001https://ieeexplore.ieee.org/document/9430766/](https://ieeexplore.ieee.org/document/9430766/)
- [58] A. Talpur and M. Gurusamy, “Machine learning for security in vehicular networks: A comprehensive survey,” *Ieee Communications Surveys and Tutorials*, vol. 24, no. 1, pp. 346–379, 2022.[Online]. Available: [〈GotoISI〉 ://WOS: 000763336500013https://ieeexplore.ieee.org/document/9619465/](https://ieeexplore.ieee.org/document/9619465/)
- [59] P. H. Rettore, A. B. Campolina, A. Souza, G. Maia, L. A. Villas, and A. A. Loureiro, “Driver authentication in vanets based on intra-vehicular sensor data,” in *2018 IEEE Symposium on Computers and Communications (ISCC)*. IEEE, 2018, pp. 00078–00083.
- [60] C.-H. Yang, D. Liang, and C.-C. Chang, “A novel driver identification method using wearables,” in *2016 13th IEEE Annual Consumer Communications & Networking Conference (CCNC)*. IEEE, 2016, pp. 1–5.
- [61] S. Shojaeilangari, W.-Y. Yau, K. Nandakumar, J. Li, and

- E. K. Teoh, Robust representation and recognition of facial emotions using extreme sparse learning, *IEEE Transactions on Image Processing*, vol. 24, no. 7, pp. 2140–2152, 2015.
- [62] K.-M. Kim and J.-I. Choi, “Passengers’ gesture recognition model in self-driving vehicles: Gesture recognition model of the passengers’ obstruction of the vision of the driver,” in *2019 IEEE 4th International Conference on Computer and Communication Systems (ICCCS)*. IEEE, Conference Proceedings, pp. 239–242.
- [63] J. Engelbrecht, M. J. Booyen, G. van Rooyen, and F. J. Bruwer, Survey of smartphone-based sensing in vehicles for intelligent transportation system applications, *IET Intelligent Transport Systems*, vol. 9, no. 10, pp. 924–935, 2015.
- [64] T. Song, X. Cheng, H. Li, J. Yu, S. Wang, and R. Bie, “Detecting driver phone calls in a moving vehicle based on voice features,” in *IEEE INFOCOM 2016–The 35th Annual IEEE International Conference on Computer Communications*. IEEE, Conference Proceedings, pp. 1–9.
- [65] J. He, J. Chen, J. Liu, and H. Li, “A lightweight architecture for driver status monitoring via convolutional neural networks,” in *2019 IEEE International Conference on Robotics and Biomimetics (ROBIO)*. IEEE, Conference Proceedings, pp. 388–394.
- [66] Y. Xing, C. Lv, Z. Zhang, H. Wang, X. Na, D. Cao, E. Velenis, and F.-Y. Wang, Identification and analysis of driver postures for in-vehicle driving activities and secondary tasks recognition, *IEEE Transactions on Computational Social Systems*, vol. 5, no. 1, pp. 95–108, 2017.
- [67] R. Chen, M. F. She, X. Sun, L. Kong, and Y. Wu, “Driver recognition based on dynamic handgrip pattern on steering wheel,” in *2011 12th ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing*. IEEE, 2011, pp. 107–112.
- [68] C. Saiprasert and S. Thajchayapong, “Remote driver identification using minimal sensory data,” *Ieee Communications Letters*, vol. 19, no. 10, pp. 1706–1709, 2015.[Online]. Available: <GotoISI> : //WOS: 000363246600011https://ieeexplore.ieee.org/document/7164248/
- [69] Z. Li, K. Zhang, B. Chen, Y. Dong, and L. Zhang, “Driver identification in intelligent vehicle systems using machine learning algorithms,” *Iet Intelligent Transport Systems*, vol. 13, no. 1, pp. 40–47, 2019.[Online]. Available: <GotoISI> : //WOS: 000454695200006
- [70] A. Chowdhury, T. Chakravarty, A. Ghose, T. Banerjee, and P. Balamuralidhar, “Investigations on driver unique identification from smartphone’s gps data alone,” *Journal of Advanced Transportation*, 2018.[Online]. Available: <GotoISI> : //WOS: 000425438900001https://downloads.hindawi.com/journals/jat/2018/9702730.pdf
- [71] S. Ezzini, I. Berrada, and M. Ghogho, “Who is behind the wheel? driver identification and fingerprinting,” *Journal of Big Data*, vol. 5, no. 1, 2018.[Online]. Available: <GotoISI> : //WOS: 000597239100001https://journalofbigdata.springeropen.com/counter/pdf/10.1186/s40537-018-0118-7.pdf
- [72] T. Tanprasert, C. Saiprasert, and S. Thajchayapong, “Combining unsupervised anomaly detection and neural networks for driver identification,” *Journal of Advanced Transportation*, 2017.[Online]. Available: <GotoISI> : //WOS:000413164300001https://downloads.hindawi.com/journals/jat/2017/6057830.pdf
- [73] W. Cao, X. Lin, K. Zhang, Y. Dong, S. Huang, and L. Zhang, “Analysis and evaluation of driving behavior recognition based on a 3-axis accelerometer using a random forest approach,” in *Proceedings of the 16th ACM/IEEE International Conference on Information Processing in Sensor Networks*, 2017, pp. 303–304.
- [74] Z. Khan, A. Koubaa, S. Fang, M. Y. Lee, and K. Muhammad, A connectivity-based clustering scheme for intelligent vehicles, *Applied Sciences*, vol. 11, no. 5, p. 2413, 2021.
- [75] F. Abbas, G. Liu, P. Fan, Z. Khan, and M. S. Bute, “A vehicle density based two-stage resource management scheme for 5g-v2x networks,” in *2020 IEEE 91st Vehicular Technology Conference (VTC2020-Spring)*. IEEE, 2020, pp. 1–5.
- [76] Abizar, H. Farman, B. Jan, Z. Khan, and A. Koubaa, A smart energy-based source location privacy preservation model for internet of things-based vehicular ad hoc networks, *Transactions on Emerging Telecommunications Technologies*, vol. 33, no. 2, p. e3973, 2022.
- [77] M. Ahmad, T. Li, Z. Khan, F. Khurshid, and M. Ahmad, A novel connectivity-based leach-meec routing protocol for mobile wireless sensor network, *Sensors*, vol. 18, no. 12, p. 4278, 2018.
- [78] S. Fang, H. Chen, Z. Khan, and P. Fan, On the content delivery efficiency of noma assisted vehicular communication networks with delay constraints, *IEEE Wireless Communications Letters*, vol. 9, no. 6, pp. 847–850, 2020.
- [79] M. A. Rahim, L. Zhu, X. Li, J. Liu, Z. Zhang, Z. Qin, S. Khan, and K. Gai, “Zero-to-stable driver identification: A non-intrusive and scalable driver identification scheme,” *Ieee Transactions on Vehicular Technology*, vol. 69, no. 1, pp. 163–171, 2020.[Online]. Available: <GotoISI> : //WOS: 000512550600014https://ieeexplore.ieee.org/document/8911519/
- [80] U. Fugiglando, E. Massaro, P. Santi, S. Milardo, K. Abida, R. Stahlmann, F. Netter, and C. Ratti, Driving behavior analysis through can bus data in an uncontrolled environment, *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, no. 2, pp. 737–748,

- 2018.
- [81] M. N. Azadani and A. Boukerche, “Convolutional and recurrent neural networks for driver identification: An empirical study,” in *NOMS 2022–2022 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2022, pp. 1–6.
- [82] L. Lu and S. Xiong, “Few-shot driver identification via meta-learning,” *Expert Systems with Applications*, vol. 203, 2022.[Online]. Available: [〈GotoISI〉 : //WOS: 000803570800002](#)
- [83] U. Ahmad, H. Song, A. Bilal, M. Alazab, and A. Jolfaei, “Securing smart vehicles from relay attacks using machine learning,” *Journal of Supercomputing*, vol. 76, no. 4, pp. 2665–2682, 2020.[Online]. Available: [〈GotoISI〉 : //WOS: 000529310600023https://link.springer.com/article/10.1007/s11227-019-03049-4](#)
- [84] M. Martínez, J. Echanobe, and I. del Campo, “Driver identification and impostor detection based on driving behavior signals,” in *2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2016, pp. 372–378.
- [85] N. Abdenmour, T. Ouni, and N. B. Amor, “Driver identification using only the can-bus vehicle data through an rcn deep learning approach,” *Robotics and Autonomous Systems*, vol. 136, 2021.[Online]. Available: [〈GotoISI〉 : //WOS: 000609454000002](#)
- [86] Y. Xun, J. Liu, N. Kato, Y. Fang, and Y. Zhang, “Automobile driver fingerprinting: A new machine learning based authentication scheme,” *Ieee Transactions on Industrial Informatics*, vol. 16, no. 2, pp. 1417–1426, 2020.[Online]. Available: [〈GotoISI〉 : //WOS: 000521337000066https://ieeexplore.ieee.org/document/8863987/](#)
- [87] F. Martinelli, F. Mercaldo, V. Nardone, and A. Santone, “Driver identification through formal methods,” *Ieee Transactions on Intelligent Transportation Systems*, vol. 23, no. 6, pp. 5625–5637, 2022.[Online]. Available: [〈GotoISI〉 : //WOS:000732171900001https://ieeexplore.ieee.org/document/9352487/](#)
- [88] J. Zhang, Z. Wu, F. Li, C. Xie, T. Ren, J. Chen, and L. Liu, “A deep learning framework for driving behavior identification on in-vehicle can-bus sensor data,” *Sensors*, vol. 19, no. 6, p. 1356, 2019.
- [89] J. Chen, Z. Wu, and J. Zhang, “Driver identification based on hidden feature extraction by using adaptive nonnegativity-constrained autoencoder,” *Applied Soft Computing*, vol. 74, pp. 1–9, 2019.[Online]. Available: [〈GotoISI〉 : //WOS: 000454251200001](#)
- [90] G. Choi, K. Lim, and S. B. Pan, “Driver identification system using 2d ecg and emg based on multistream cnn for intelligent vehicle,” *Ieee Sensors Letters*, vol. 6, no. 6, 2022.[Online]. Available: [〈GotoISI〉 : //WOS: 000809709800001https://ieeexplore.ieee.org/document/9782761/](#)
- [91] Z. Li, H. Wang, Y. Zhang, and X. Zhao, “Random forest-based feature selection and detection method for drunk driving recognition,” *International Journal of Distributed Sensor Networks*, vol. 16, no. 2, p. 1550147720905234, 2020.
- [92] X. Zhang, X. Zhao, and J. Rong, “A study of individual characteristics of driving behavior based on hidden markov model,” *Sensors and Transducers*, vol. 167, no. 3, p. 194, 2014.
- [93] T. Wakita, K. Ozawa, C. Miyajima, K. Igarashi, K. Itou, K. Takeda, and F. Itakura, “Driver identification using driving behavior signals,” *IEICE TRANSACTIONS on Information and Systems*, vol. 89, no. 3, pp. 1188–1194, 2006.
- [94] C. Miyajima, Y. Nishiwaki, K. Ozawa, T. Wakita, K. Itou, K. Takeda, and F. Itakura, “Driver modeling based on driving behavior and its evaluation in driver identification,” *Proceedings of the IEEE*, vol. 95, no. 2, pp. 427–437, 2007.
- [95] A. Burton, T. Parikh, S. Mascarenhas, J. Zhang, J. Voris, N. S. Artan, and W. Li, “Driver identification and authentication with active behavior modeling,” in *2016 12th International Conference on Network and Service Management (CNSM)*. IEEE, Conference Proceedings, pp. 388–393.
- [96] H. Cai, Z. Hu, Z. Chen, and D. Zhu, “A driving fingerprint map method of driving characteristic representation for driver identification,” *Ieee Access*, vol. 6, pp. 71012–71019, 2018.[Online]. Available: [〈GotoISI〉 : //WOS: 000453303400001https://ieeexplore.ieee.org/ielx7/6287639/8274985/08537901.pdf?tp=&arnumber=8537901&isnumber=8274985&ref=](#)
- [97] K. S. Ahmad, A. S. Thosar, J. H. Nirmal, and V. S. Pande, “A unique approach in text independent speaker recognition using mfcc feature sets and probabilistic neural network,” in *2015 Eighth International Conference on Advances in Pattern Recognition (ICAPR)*. IEEE, 2015, pp. 1–6.
- [98] H. Ali, S. N. Tran, E. Benetos, and A. S. d’Avila Garcez, “Speaker recognition with hybrid features from a deep belief network,” *Neural Computing and Applications*, vol. 29, pp. 13–19, 2018.
- [99] D. A. Reynolds and R. C. Rose, “Robust text-independent speaker identification using gaussian mixture speaker models,” *IEEE transactions on speech and audio processing*, vol. 3, no. 1, pp. 72–83, 1995.
- [100] R. Jahangir, Y. W. Teh, H. F. Nweke, G. Mujtaba, M. A. Al-Garadi, and I. Ali, “Speaker identification through artificial intelligence techniques: A comprehensive review and research challenges,” *Expert Systems with Applications*, vol. 171, p. 114591, 2021.
- [101] T. F. Covões and E. R. Hruschka, “Unsupervised learning of gaussian mixture models: Evolutionary create and eliminate for expectation maximization algorithm,” in *2013 IEEE Congress on Evolutionary Computation*. IEEE, 2013, pp. 3206–3213.
- [102] G. Tian, Y. Xia, Y. Zhang, and D. Feng, “Hybrid genetic and variational expectation-maximization algorithm for

- gaussian-mixture-model-based brain mr image segmentation, *IEEE transactions on information technology in biomedicine*, vol. 15, no. 3, pp. 373–380, 2011.
- [103] H. Qian, Y. Ou, X. Wu, X. Meng, Y. Xu et al., “Support vector machine for behavior-based driver identification system,” *Journal of Robotics*, vol. 2010, 2010.
- [104] N. Mazhar, R. Salleh, M. Zeeshan, M. M. Hameed, and N. Khan, “R-idps: Real time sdn based idps system for iot security,” in *2021 IEEE 18th International Conference on Smart Communities: Improving Quality of Life Using ICT, IoT and AI (HONET)*. IEEE, 2021, pp. 71–76.
- [105] C. Cortes and V. Vapnik, Support-vector networks, *Machine learning*, vol. 20, pp. 273–297, 1995.
- [106] R. Abbey, T. He, and T. Wang, “Methods of multinomial classification using support vector machines,” *Paper SAS434-2017*.
- [107] C. Li, R.-V. Sanchez, G. Zurita, M. Cerrada, D. Cabrera, and R. E. Vásquez, Multimodal deep support vector classification with homologous features and its application to gearbox fault diagnosis, *Neurocomputing*, vol. 168, pp. 119–127, 2015.
- [108] J. A. Suykens and J. Vandewalle, Least squares support vector machine classifiers, *Neural processing letters*, vol. 9, pp. 293–300, 1999.
- [109] H. Qian, Y. Ou, X. Wu, X. Meng, and Y. Xu, “Support vector machine for behavior-based driver identification system,” *Journal of Robotics*, vol. 2010, 2010.
- [110] Z. Li, H. Wang, Y. Zhang, and X. Zhao, Random forest-based feature selection and detection method for drunk driving recognition, *International Journal of Distributed Sensor Networks*, vol. 16, no. 2, p. 1550147720905234, 2020.
- [111] J.-D. Wu and S.-H. Ye, Driver identification using fingervein patterns with radon transform and neural network, *Expert Systems with Applications*, vol. 36, no. 3, pp. 5793–5799, 2009.
- [112] W. Dong, J. Li, R. Yao, C. Li, T. Yuan, and L. Wang, “Characterizing driving styles with deep learning,” *arXiv preprint arXiv: 1607.03611*, 2016.
- [113] D. Jeong, M. Kim, K. Kim, T. Kim, J. Jin, C. Lee, and S. Lim, “Real-time driver identification using vehicular big data and deep learning,” in *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2018, pp. 123–130.
- [114] W. G. Hatcher and W. Yu, A survey of deep learning: Platforms, applications and emerging research trends, *IEEE Access*, vol. 6, pp. 24411–24432, 2018.
- [115] N. Khan, R. bin Salleh, Z. Khan, A. Koubaa, M. Hamdan, and A. M. Abdelmoniem, Ensuring reliable network operations and maintenance: The role of pmrf for switch maintenance and upgrades in sdn, *Journal of King Saud University-Computer and Information Sciences*, vol. 35, no. 10, p. 101809, 2023.
- [116] S. Ullah, W. Boulila, A. Koubaa, and J. Ahmad, “Magruids: A multi-head attention-based gated recurrent unit for intrusion detection in iiot networks,” *IEEE Access*, 2023.
- [117] S. Ullah, J. Ahmad, M. A. Khan, M. S. Alshehri, W. Boulila, A. Koubaa, S. U. Jan, and M. M. I. Ch, Tnnids: Transformer neural network-based intrusion detection system for mqtt-enabled iot networks, *Computer Networks*, vol. 237, p. 110072, 2023.
- [118] M. H. Jamal, M. A. Khan, S. Ullah, M. S. Alshehri, S. Almakdi, U. Rashid, A. Alazeb, and J. Ahmad, Multistep attack detection in industrial networks using a hybrid deep learning architecture, *Mathematical Biosciences and Engineering*, vol. 20, no. 8, pp. 13824–13848, 2023.
- [119] M. A. Jan, X. He, H. Song, and M. Babar, Machine learning and big data analytics for iot-enabled smart cities, *Mobile Networks and Applications*, vol. 26, pp. 156–158, 2021.
- [120] S. Mozaffari, O. Y. Al-Jarrah, M. Dianati, P. Jennings, and A. Mouzakitis, Deep learning-based vehicle behavior prediction for autonomous driving applications: A review, *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 1, pp. 33–47, 2020.
- [121] G. Tesauro, “Practical issues in temporal difference learning,” *Advances in neural information processing systems*, vol. 4, 1991.
- [122] M. M. Bejani and M. Ghatee, Convolutional neural network with adaptive regularization to classify driving styles on smartphones, *IEEE transactions on intelligent transportation systems*, vol. 21, no. 2, pp. 543–552, 2019.
- [123] A. Krizhevsky, I. Sutskever, and G. E. Hinton, Imagenet classification with deep convolutional neural networks, *Communications of the ACM*, vol. 60, no. 6, pp. 84–90, 2017.
- [124] J. He, J. Chen, J. Liu, and H. Li, “A lightweight architecture for driver status monitoring via convolutional neural networks,” in *2019 IEEE International Conference on Robotics and Biomimetics (ROBIO)*. IEEE, 2019, pp. 388–394.
- [125] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, Gradientbased learning applied to document recognition, *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [126] A. Girma, X. Yan, and A. Homaifar, “Driver identification based on vehicle telematics data using lstm-recurrent neural network,” in *2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI)*. IEEE, 2019, pp. 894–902.
- [127] W. Dong, T. Yuan, K. Yang, C. Li, and S. Zhang, “Autoencoder regularized network for driving style representation learning,” *arXiv preprint arXiv: 1701.01272*, 2017.
- [128] A. Koubaa, W. Boulila, L. Ghouti, A. Alzahem, and S. Latif, “Exploring chatgpt capabilities and limitations: A survey,” *IEEE Access*, 2023.
- [129] H. Ali, The potential of gpt-4 as a personalized virtual assistant for bariatric surgery patients, *Obesity Surgery*, vol. 33, no. 5, pp. 1605–1605, 2023.
- [130] L. C. Leonard, Web-based behavioral modeling for continuous user authentication (CUA), *Elsevier*, vol. 1, pp. 105–44, 2017.



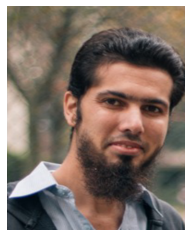
Anwar Mehmood Sohail received the B.S. degree in Electrical Computer Engineering with distinction from the COMSATS Univerisy Islamabad, Abbottabad, Pakistan, in 2014, and the M.S. degree in Computer Systems Engineering from University of Engineering and Technology (UET)

Peshawar in 2019. He is currently a Research Assistant with the National Centre for Big Data and Cloud Computing (NCBC), UET Peshawar, and pursuing the Ph.D. degree in Computer Science at the University of Malaya, Malaysia. His research interests include machine learning, deep learning, natural language processing, and intelligent transportation systems.



Teh Ying Wah is a highly accomplished computer scientist and data mining expert with over 35 years of experience, he demonstrated exceptional leadership, expertise, and vision in the field. Throughout his career, he achieved numerous successes and significantly contributed to the industry. He began as an

entry-level computer programmer in 1988 and advanced to become a Professor of Data Mining at the Faculty of Computer Science and Information Technology at the University of Malaya. He obtained his tertiary academic qualifications from Oklahoma City University and the University of Malaya, and he published more than 90 academic papers in top-tier journals, including Information Fusion and the International Journal of Information Management. He has a remarkable H-index and number of citations in Web of Science, Scopus, and Google Scholar databases, and he supervised numerous students at all levels of study. His areas of research include data warehouse, data mining, deep learning, IoT, activity recognition, wearable sensors, accelerometers, heart arrhythmia, electrocardiograph, supraventricular premature beat, multivariate time series, edge computing, task scheduling, data streams, mobile computing, speaker verification, language recognition, clustering algorithms, MapReduce, stock market, and sentiment analysis.



Nauman Khan received his Master's degree from the University of Electronic Science and Technology of China in 2016 and received his Ph.D. in Computer Science from Universiti Malaya, Malaysia. He is currently serving as a senior lecturer at the University of Malakand, Pakistan, and a Remote Research Fellow with the

Robotics and Internet-of-Things (RIOTU) Laboratory at Prince Sultan University, Saudi Arabia. He is a certified Cisco and Microsoft professional with prior experience as a Cisco-certified network engineer in both national and international organizations. His research interests include software-defined networking (SDN), network security, the Internet of Things (IoT), machine learning, and large language models (LLMs). He has published several research articles in top-tier journals and serves as a reviewer for leading venues, including the IEEE Communications Surveys& Tutorials and flagship conferences such as IEEE VTS.



Zahid Khan (IEEE SM'23) is currently a senior researcher and leads the Intelligent Transportation Systems (ITS) section of the Robotics and Internet-of-Things Laboratory (RIOTU) at Prince Sultan University, Saudi Arabia. He received his Ph.D. in Information and Communication Engineering from Southwest Jiaotong University, Chengdu, China, in 2019, and a Master's degree in Computer Science from the University of Nice Sophia Antipolis, Nice, France, in 2015. Additionally, he completed a Postdoctoral Fellowship in Intelligent Transportation Systems at RIOTU, Prince Sultan University, Saudi Arabia. Dr. Zahid has published over 50 research articles in renowned journals and conferences, including IEEE Transactions and other flagship conferences. Beyond his research contributions, he is actively engaged in editorial services and academic leadership, having served extensively as a reviewer, guest editor, and conference chair. He plays a vital role in advancing the scholarly community by organizing and leading high-impact academic events. Dr. Zahid's research interests lie in Internet of Things (IoT) applications within intelligent transportation systems, focusing on cluster-based routing techniques to improve link reliability and connectivity in Vehicular Ad-hoc Networks (VANETs).



Prof. Anis Koubaa is a Full Professor of Computer Science and Software Engineering at Alfaisal University. Professor Anis Koubaa is a distinguished leader in Artificial Intelligence, Robotics, and Computer Science. He served as the Director of the Research and Initiatives Center and the Innovation Center at PSU,

and has held pivotal roles such as Aide to the President for Research Governance, significantly enhancing the university's global research impact. With over two decades of groundbreaking contributions, Professor Koubaa has advanced fields including Generative AI, Robotics, Drones, and the Internet of Things (IoT), with impactful applications in smart cities, healthcare, education, and business transformation. He is renowned for developing the Arabic Large Language Models (ArabianLLM) series and leading innovative AI projects in surveillance, computer vision, and automation. A Senior Fellow of the Higher Education Academy (UK) and editor of over 20 Springer publications, Professor Koubaa's exceptional work has earned him global recognition. Among his prestigious accolades are:

- The Shoman Award for Arab Researchers (2024) for excellence in AI and robotics.
- Third Prize in the SDAIA ALLAM Competition (2024), selected from over 170 international teams for his contributions to AI innovation.
- Numerous other awards, including the AI Leadership Award (2022) and the KAUST Challenge Award (2021).

Recognized among the career-based top 2% scientists by Stanford University, Professor Koubaa continues to lead transformative research and development initiatives. His expertise in AI-driven business transformation, smart logistics, and cloud robotics positions him as a pioneer at the forefront of global AI and technological innovation